Resilient Communication Networks

Fundamentals of Communication Networks



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Disclaimer: some of these slides have been derived from Günter Schäfer's lecture „Protection of Communication Infrastructures", Mathias Fischers Lecture „Resilient Networks", Matthias Hollicks Lecture „ Ringvorlesung KRITIS", and Ralf Steinmetz's lecture Communication Networks II"

Key to Internet Success: Layers

Applications

… built on…

Reliable (or unreliable) transport

… built on…

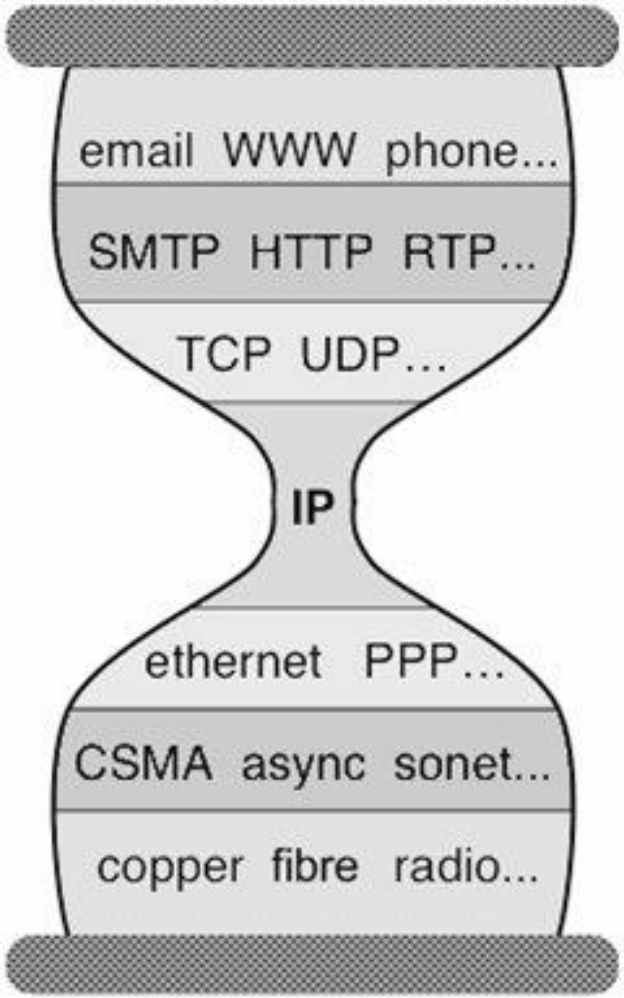
Best-effort global packet delivery

… built on…

Best-effort local packet delivery

… built on…

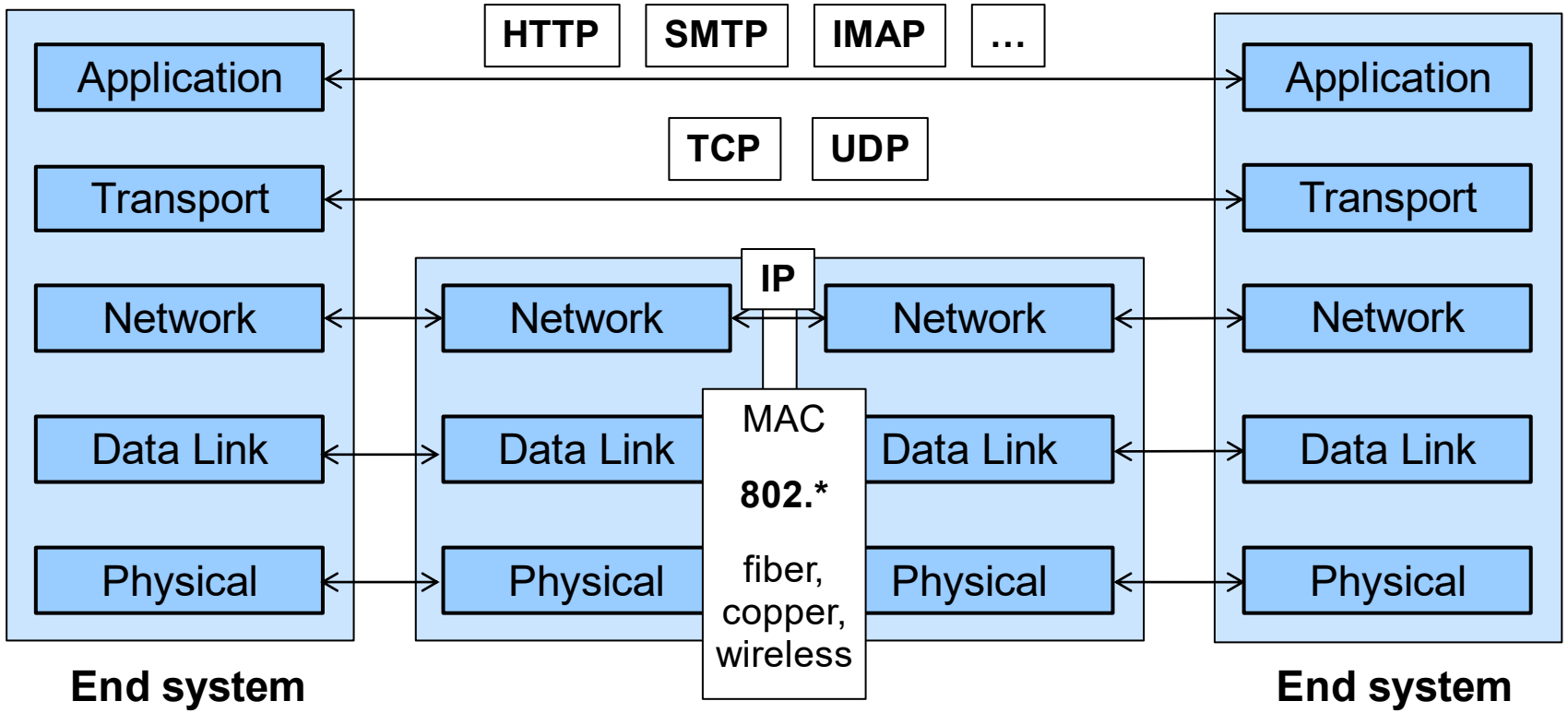
Physical transfer of bits



The Internet Protocol Stack: A Success Story

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The TCP/IP and ISO/OSI layers remained the same for 40+ years



What happens when you connect your laptop to a network?

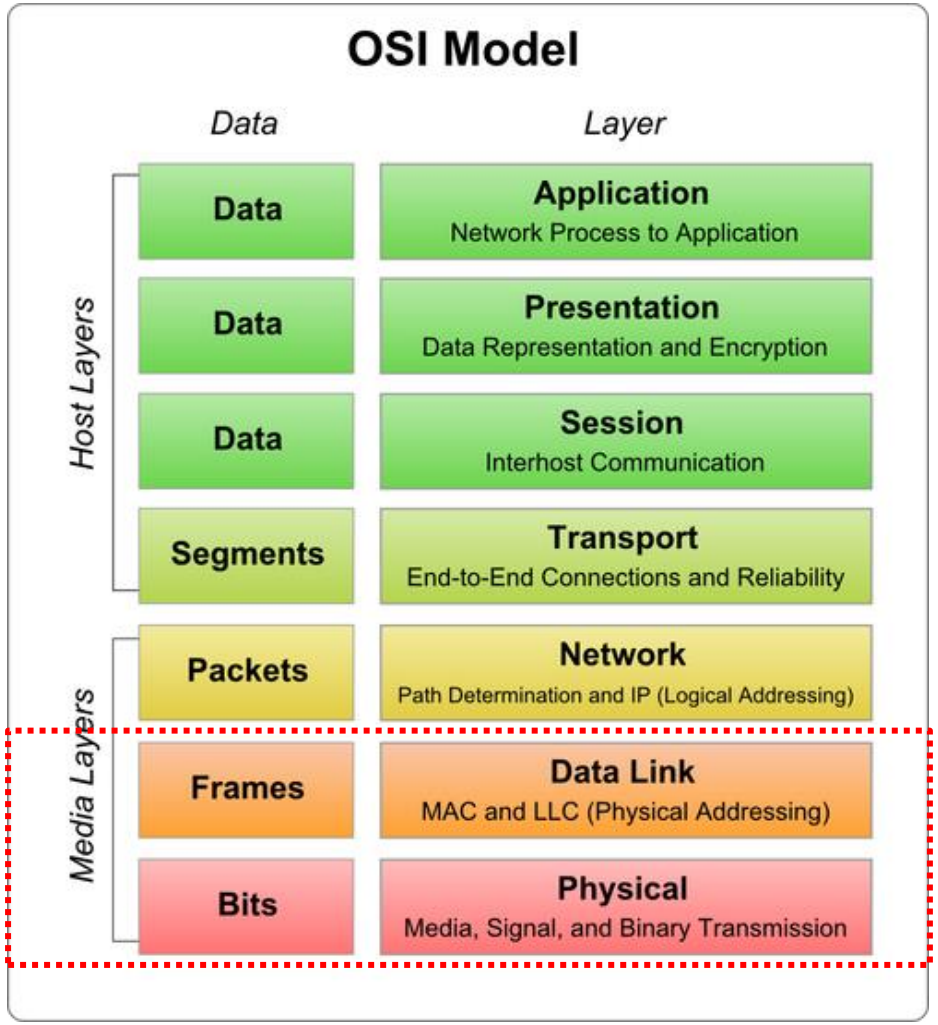
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You first establish a connection on Layer 1+2

- Ethernet
- Wifi
- Cellular

On layer 2, each device has one or more unique Medium Access Control (MAC) addresses

- 48bit identifier
- Often represented using hexadecimal values: `00:80:41:ae:fd:7e`
- Are used for addressing packets inside a layer 2 network
- Must be unique in the network



What happens when you connect your laptop to a network?

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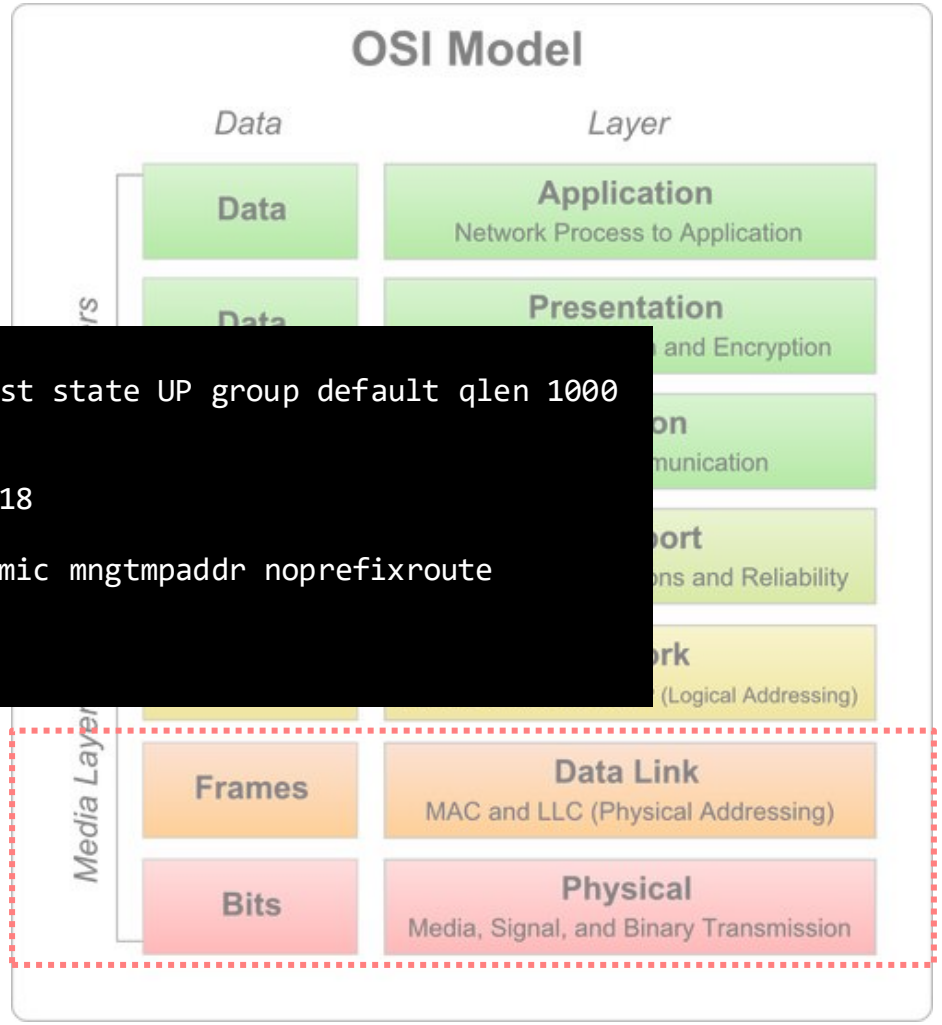
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- Ethernet
- Wifi
- Cellular

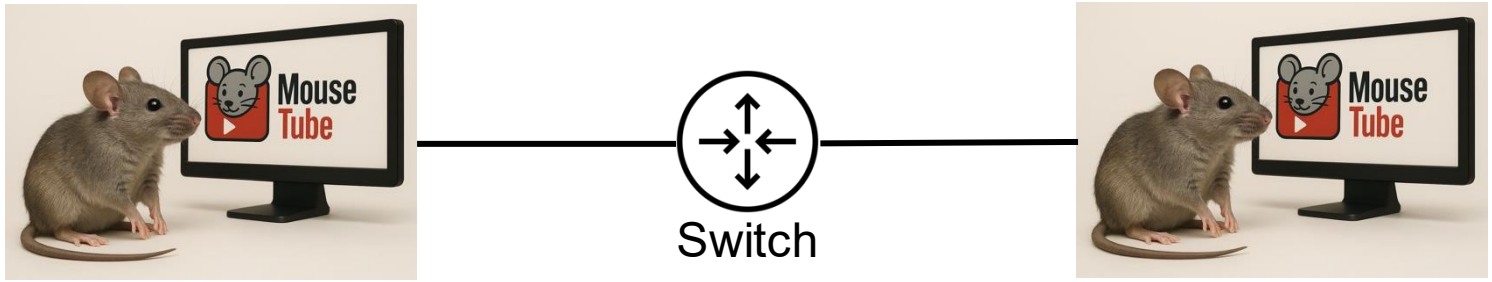
```
ubuntu@ubuntu:~$ ip a
2: ens18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether cd:11:98:fc:2d:32 brd ff:ff:ff:ff:ff:ff
    altname enp0s18
    inet 10.2.2.184/16 metric 100 brd 10.2.255.255 scope global ens18
       valid_lft forever preferred_lft forever
    inet6 2001:41b8:810:41:be24:11ff:feff:1f16/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86260sec preferred_lft 14260sec
    inet6 fe80::be24:11ff:feff:1f16/64 scope link
       valid_lft forever preferred_lft forever
```

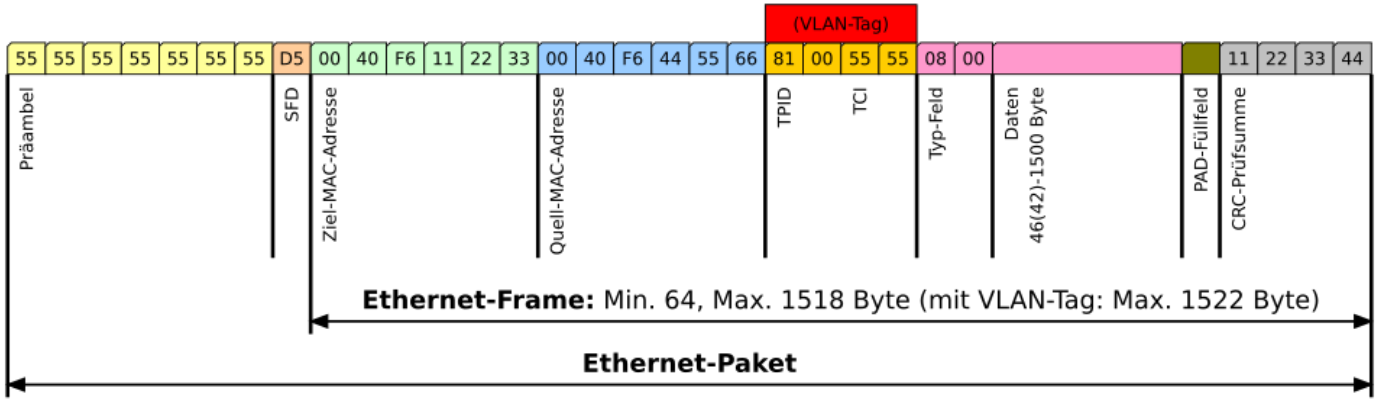
On layer 2 Access Co

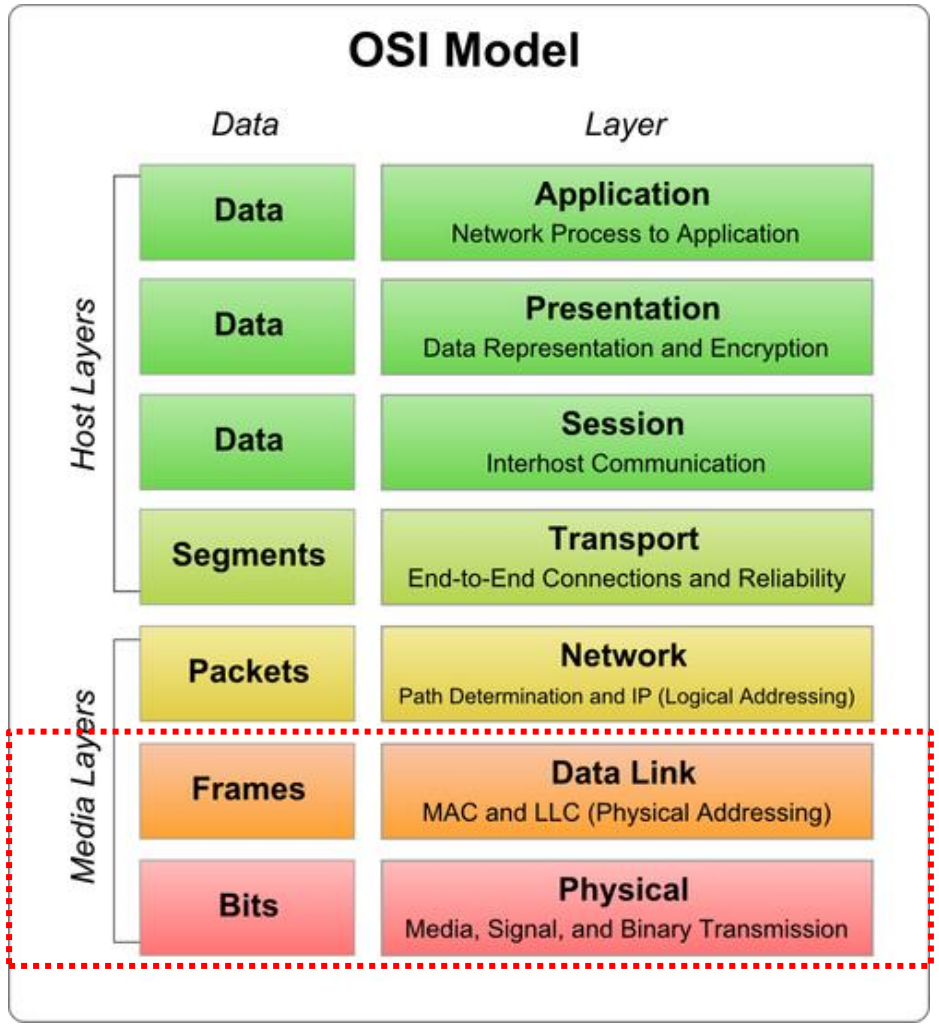
- 48bit id
- Often re
  00:80:
- Are used for addressing packets inside a layer 2 network



Ethernet


Mouse
Tube
Switch
Mouse
Tube

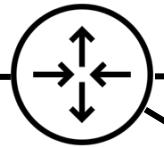

(VLAN-Tag)
55 55 55 55 55 55 55 D5 00 40 F6 11 22 33 00 40 F6 44 55 66 81 00 55 55 08 00 11 22 33 44
Präambel
SFD
Ziel-MAC-Adresse
Quell-MAC-Adresse
TPID
TCI
Typ-Feld
Daten
46(42)-1500 Byte
PAD-Füllfeld
CRC-Prüfsumme
Ethernet-Frame: Min. 64, Max. 1518 Byte (mit VLAN-Tag: Max. 1522 Byte)
Ethernet-Paket


OSI Model
Data
Layer
Host Layers
Media Layers
Data
Application
Network Process to Application
Data
Presentation
Data Representation and Encryption
Data
Session
Interhost Communication
Segments
Transport
End-to-End Connections and Reliability
Packets
Network
Path Determination and IP (Logical Addressing)
Frames
Data Link
MAC and LLC (Physical Addressing)
Bits
Physical
Media, Signal, and Binary Transmission

Ethernet
Unicast Frames



C0:FF:EE:00:BA:BE





C0:DE:C0:DE:C0:DE



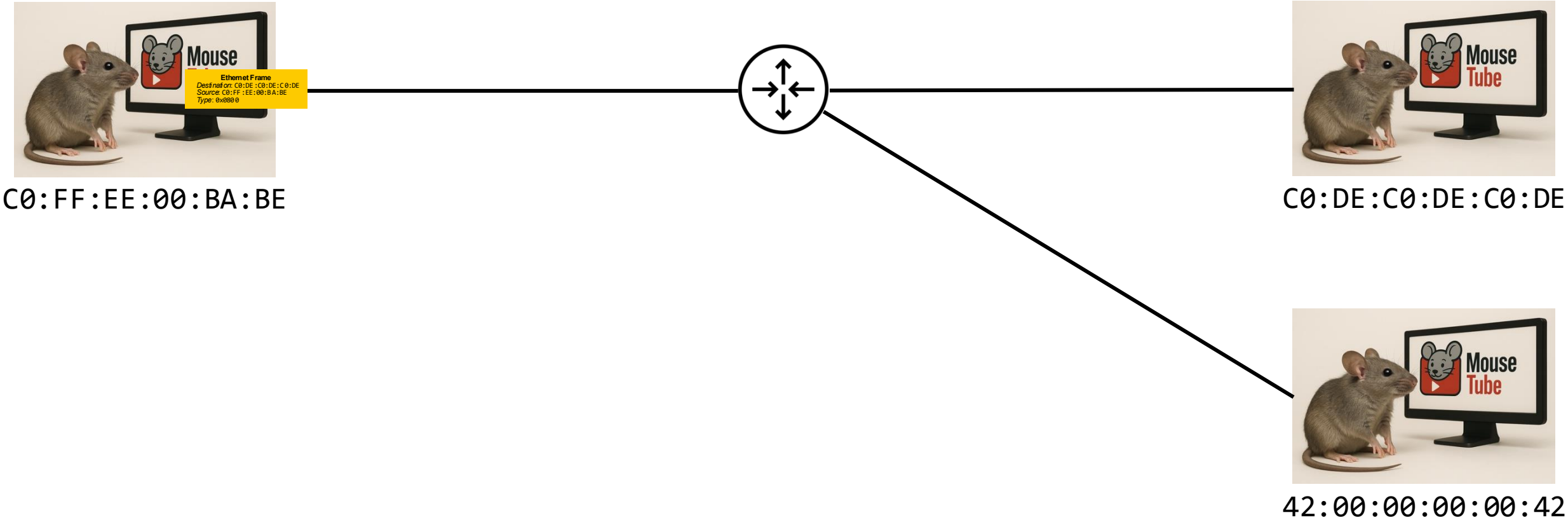
42:00:00:00:00:42

Ethernet Frame
Destination: C0:DE:C0:DE:C0:DE
Source: C0:FF:EE:00:BA:BE
Type: 0x0800

MAC addresses of devices

IPv4

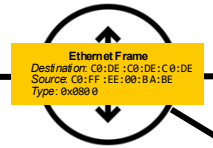
Ethernet
Unicast Frames



Ethernet
Unicast Frames



C0:FF:EE:00:BA:BE





C0:DE:C0:DE:C0:DE



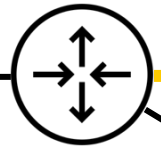
42:00:00:00:00:42

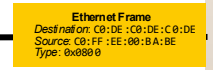
What happens with the unicast ethernet frame?

Ethernet
Unicast Frames



C0:FF:EE:00:BA:BE







C0:DE:C0:DE:C0:DE



42:00:00:00:00:42

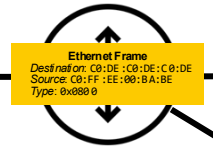
What happens with the unicast ethernet frame?

1. If the destination MAC address is known, the frame is only sent on the port where the respective device is connected

Ethernet
Unicast Frames



C0:FF:EE:00:BA:BE





C0:DE:C0:DE:C0:DE



42:00:00:00:00:42

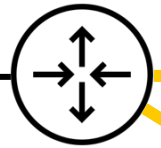
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Ethernet
Unicast Frames



C0:FF:EE:00:BA:BE





C0:DE:C0:DE:C0:DE



42:00:00:00:00:42

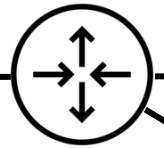
What happens with the unicast ethernet frame?

1. If the destination MAC address is known, the frame is only sent on the port where the respective device is connected
2. If the destination MAC is not known, the frame is broadcasted to all connected devices

Ethernet
Broadcast Frames



C0:FF:EE:00:BA:BE





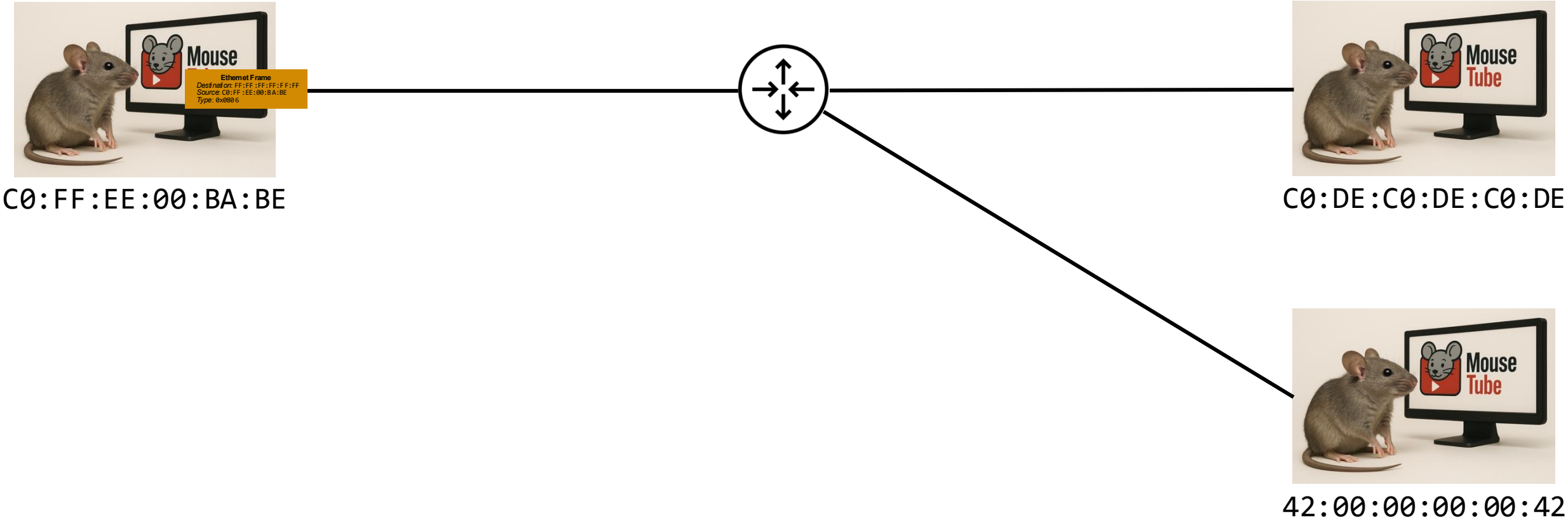
C0:DE:C0:DE:C0:DE



42:00:00:00:00:42

Ethernet Frame
Destination: FF:FF:FF:FF:FF:FF ← Broadcast address
Source: C0:FF:EE:00:BA:BE ← MAC address of device
Type: 0x0806 ← ARP

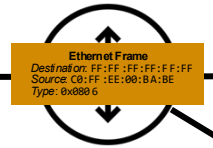
Ethernet
Broadcast Frames



Ethernet
Broadcast Frames



C0:FF:EE:00:BA:BE





C0:DE:C0:DE:C0:DE



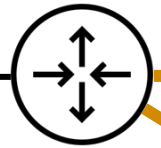
42:00:00:00:00:42

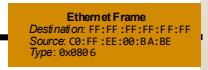
What happens with the broadcast ethernet frame?

Ethernet
Broadcast Frames



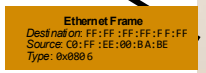
C0:FF:EE:00:BA:BE







C0:DE:C0:DE:C0:DE





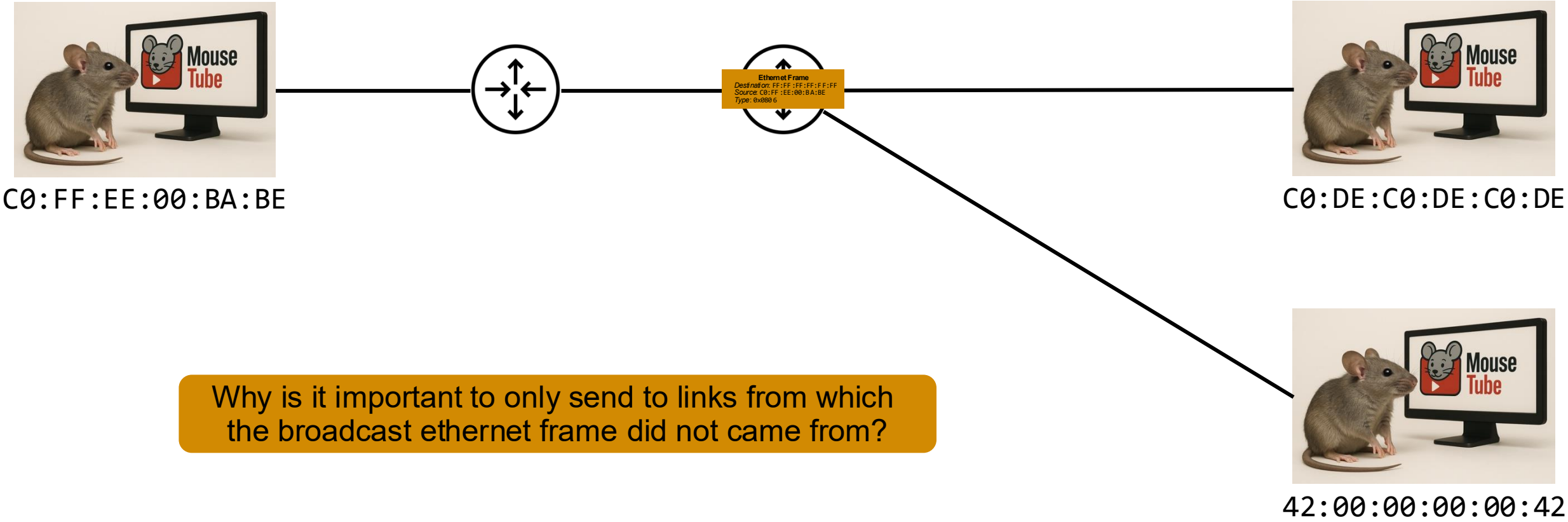
42:00:00:00:00:42

What happens with the broadcast ethernet frame?

The packet is sent via all links except the one where it came from

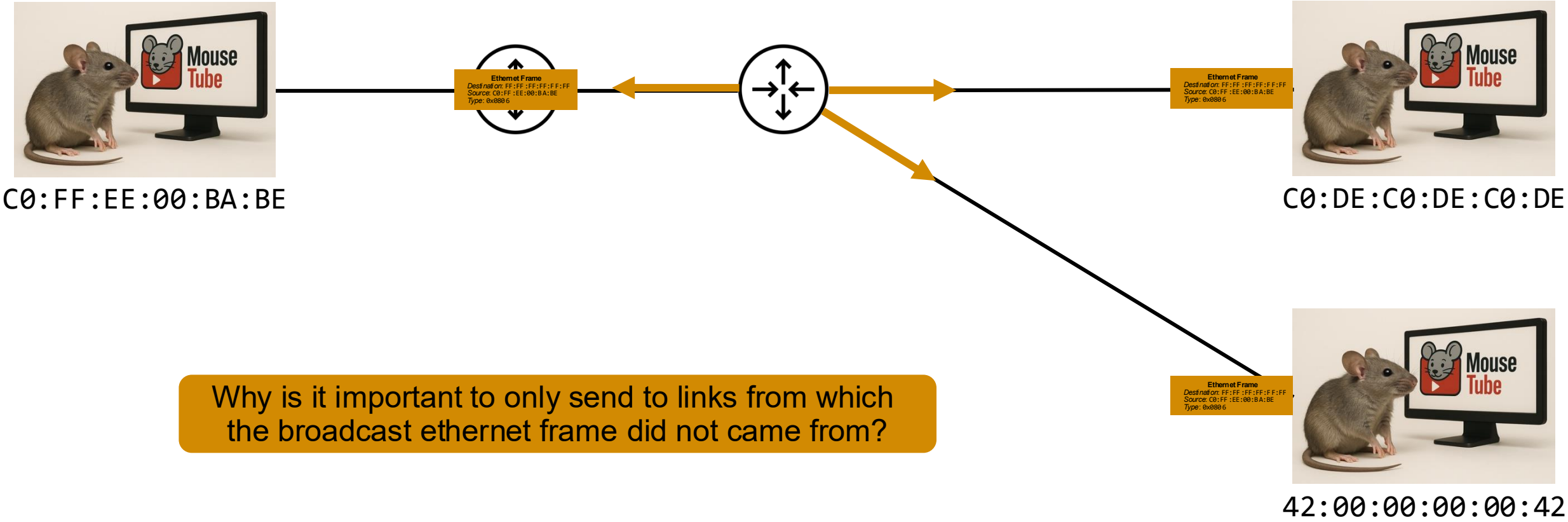
Why is it important to only send to links from which the broadcast ethernet frame did not came from?

Ethernet
Broadcast Frames



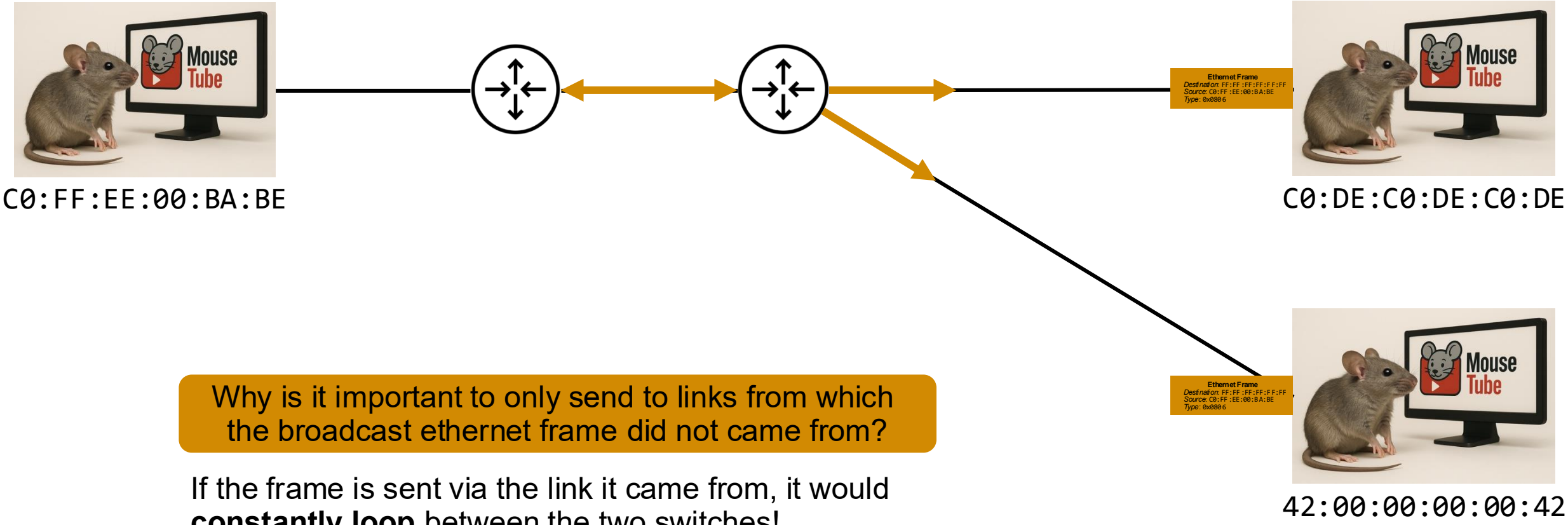
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Ethernet
Broadcast Frames



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Ethernet
Broadcast Frames



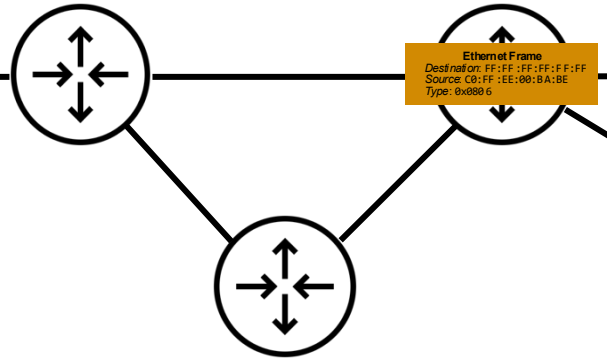
Why is it important to only send to links from which the broadcast ethernet frame did not came from?

If the frame is sent via the link it came from, it would **constantly loop** between the two switches!

Ethernet
Loops



C0:FF:EE:00:BA:BE





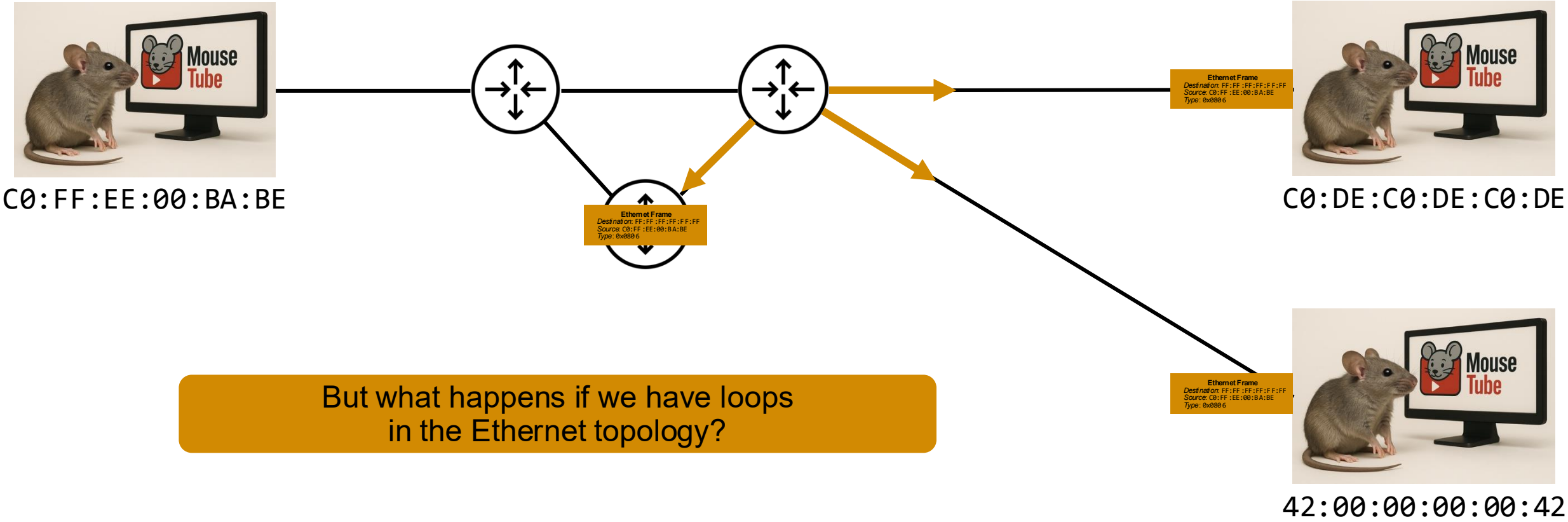
C0:DE:C0:DE:C0:DE



42:00:00:00:00:42

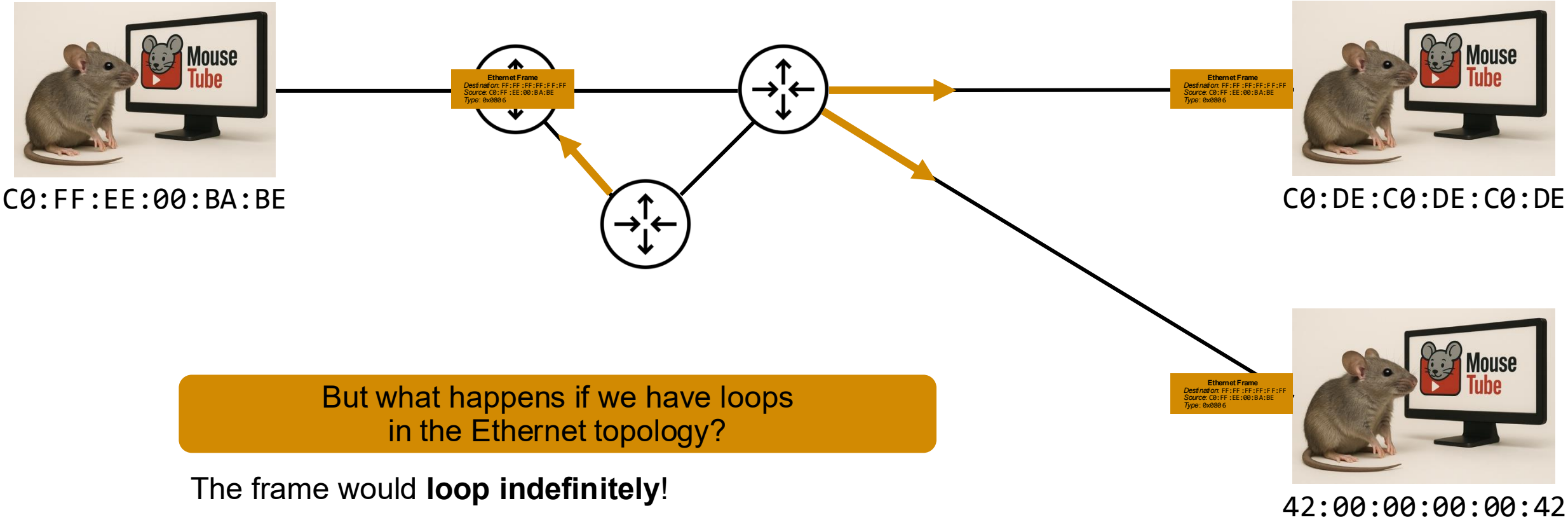
But what happens if we have loops
in the Ethernet topology?

Ethernet
Loops



But what happens if we have loops in the Ethernet topology?

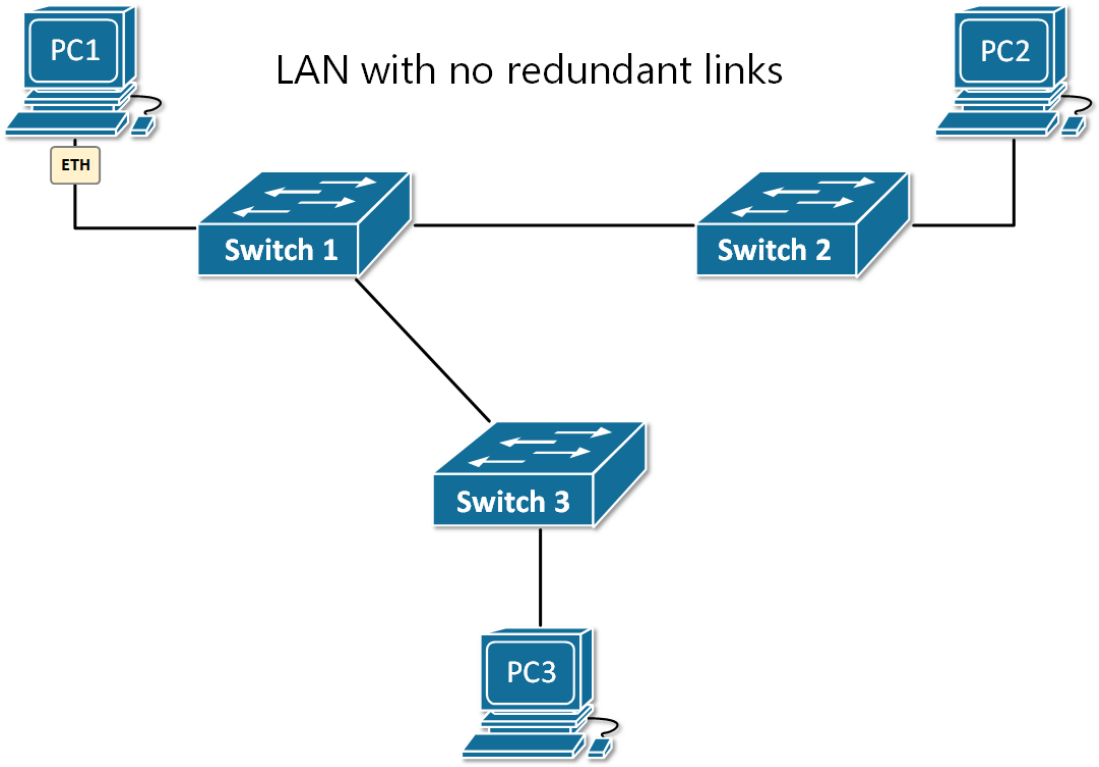
Ethernet
Loops

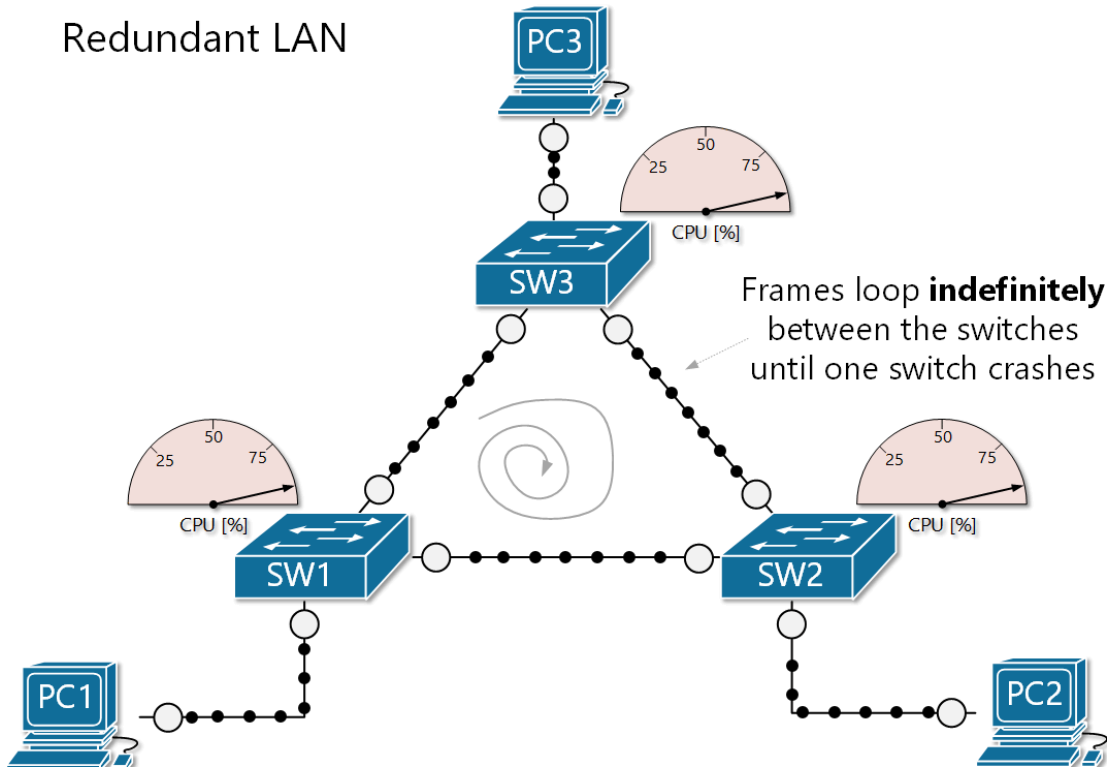


But what happens if we have loops in the Ethernet topology?

The frame would **loop indefinitely**!

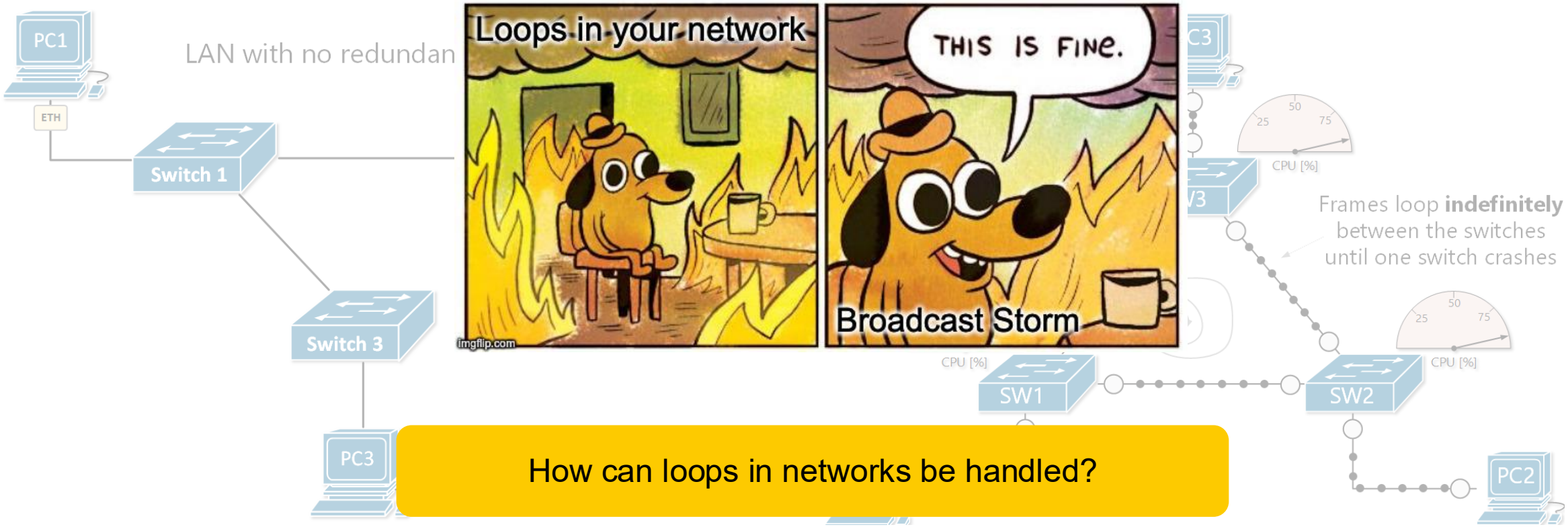
Ethernet
Loops





https://www.networkacademy.io/ccna/spanning-tree/why-do-we-need-stp

Ethernet
Loops



How can loops in networks be handled?

https://www.networkacademy.io/ccna/spanning-tree/why-do-we-need-stp

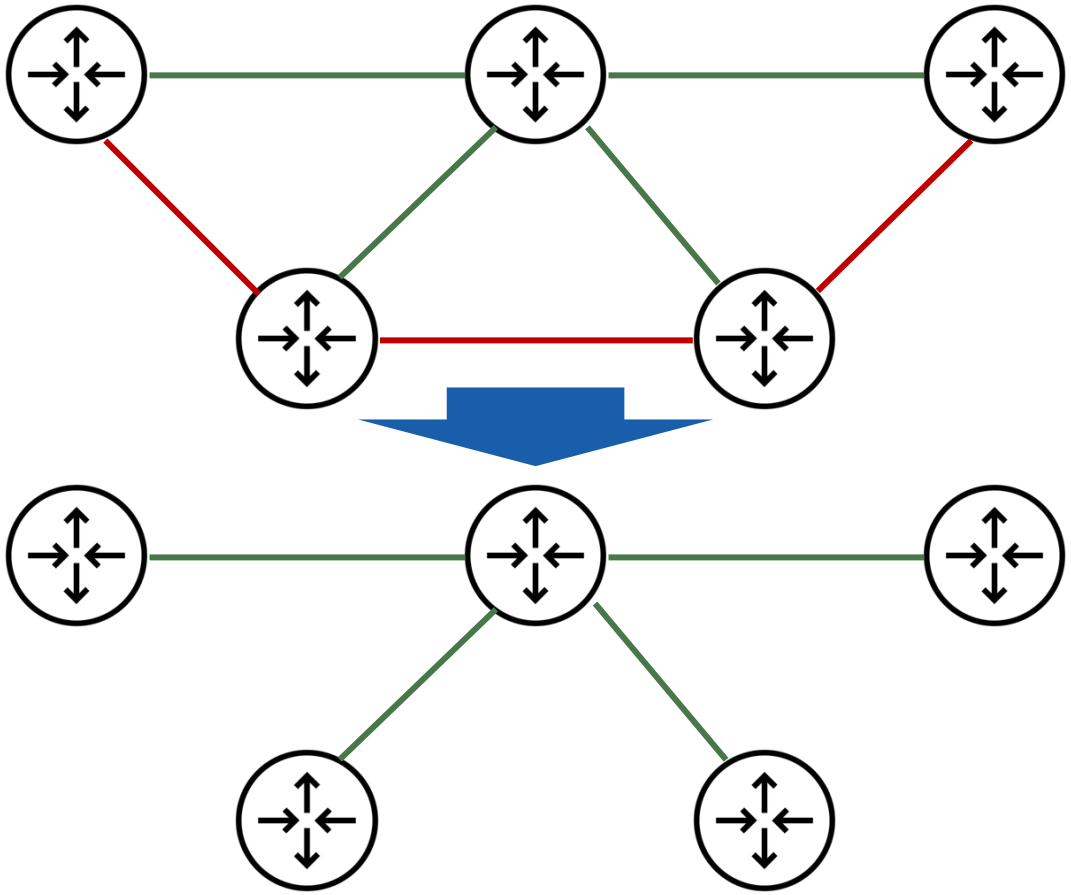
In the simplest approach, loops are just removed manually

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Especially in small networks, you can ensure that no loops exist

- Remove any links that cause a loop (either by deactivating or removing the cable)
- Disadvantage: Huge manual effort, little resilience

Mechanisms (STP, ERPS, …) are available to dynamically adjust the topology to avoid loops in networks



Why do I need another layer?

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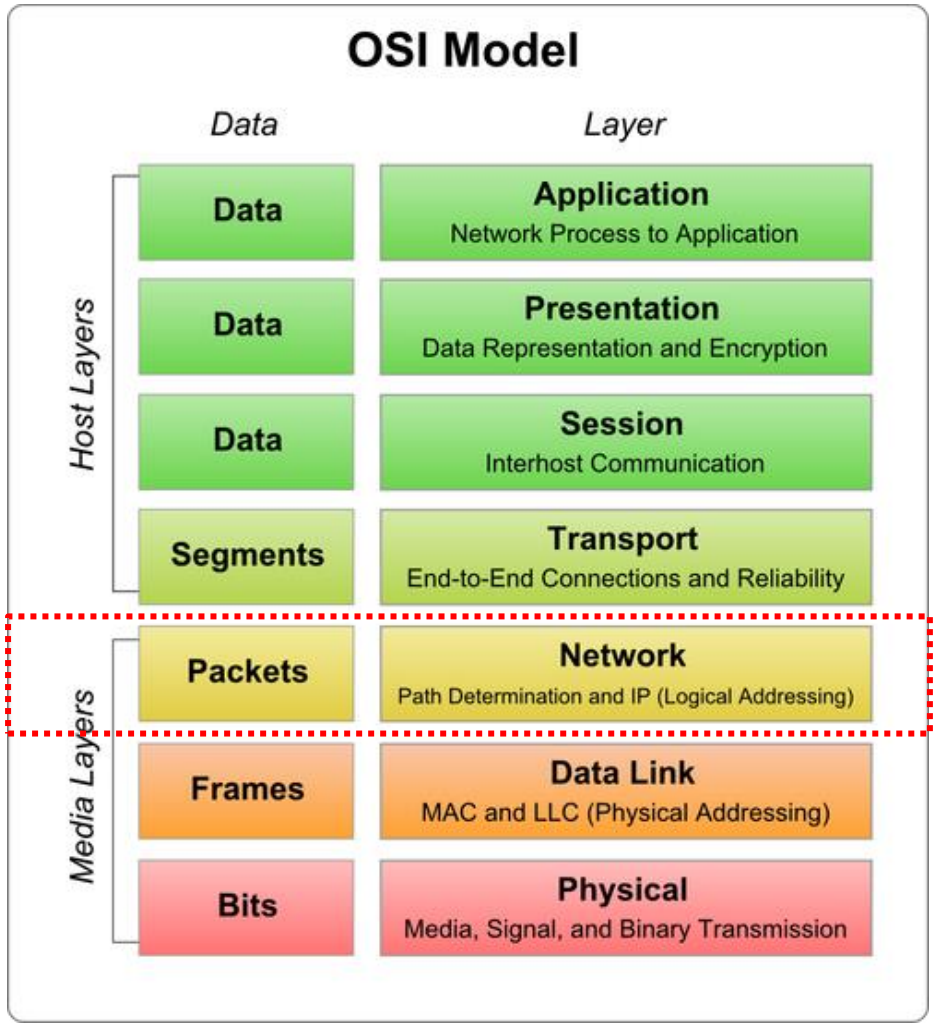
On layer 2, a lot of communication is based on broadcast

- Device discovery
- Communication to unknown devices

Failures have a large impact on the network

- A malfunctioning device can disrupt the whole network
- Loops are (not directly) possible on layer 2
- All devices may talk to all other devices
- All switches need to know the location of all devices in the network

On the network layer, basically all communication is unicast and often hierarchically organized



What happens on layer 3?

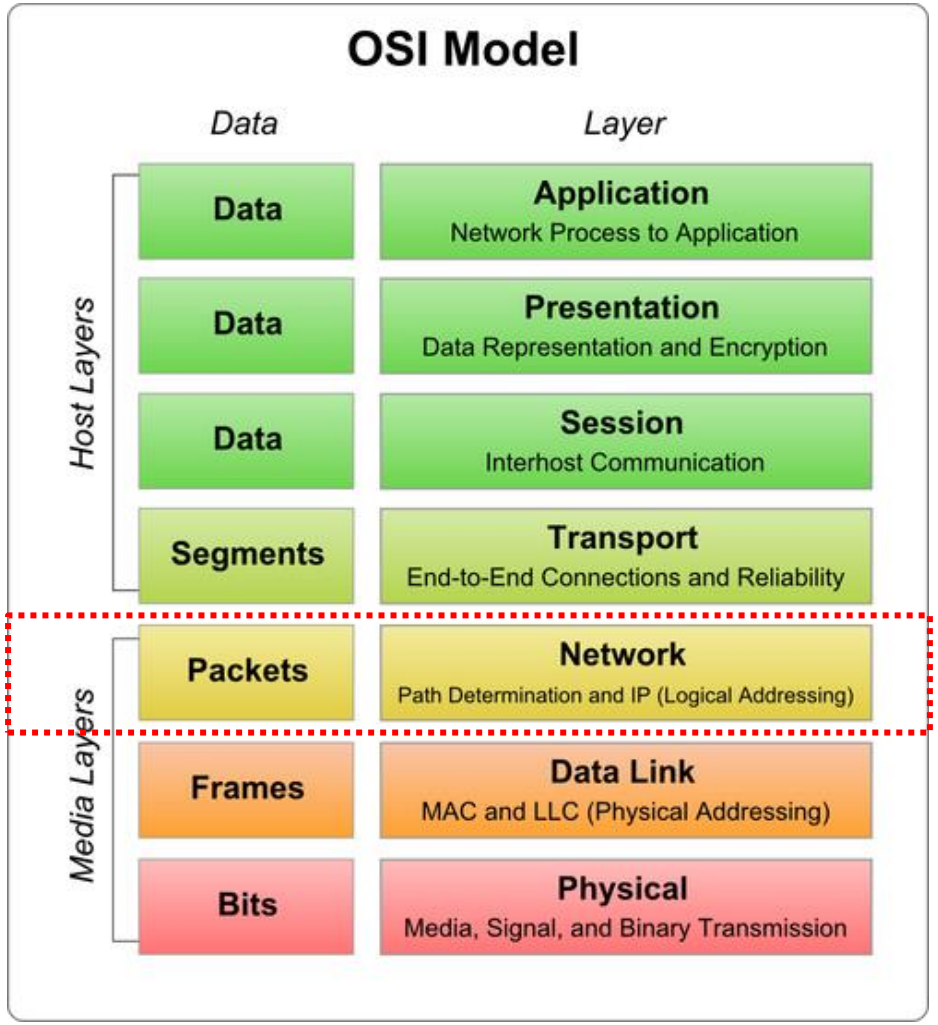
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Once a device is connected on layer 2, it tries to obtain a valid address for layer 3

- Configured on the host device
- Via Dynamic Host Configuration Protocol (DHCP)

The layer 3 address is, in essence, used to communicate with all other devices in the network

- IPv4 or IPv6
- IPv4 address (32bit, often in decimal form): `192.168.1.15`
- IPv6 address (128bit, often in hexademical form): `2001:0db8:85a3:08d3:1319:8a2e:0370:7344`
- Ignoring functionalities like Network Access Translation (NAT) for now



Dynamic Host Configuration Protocol (DHCP)

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Manages addresses in a network

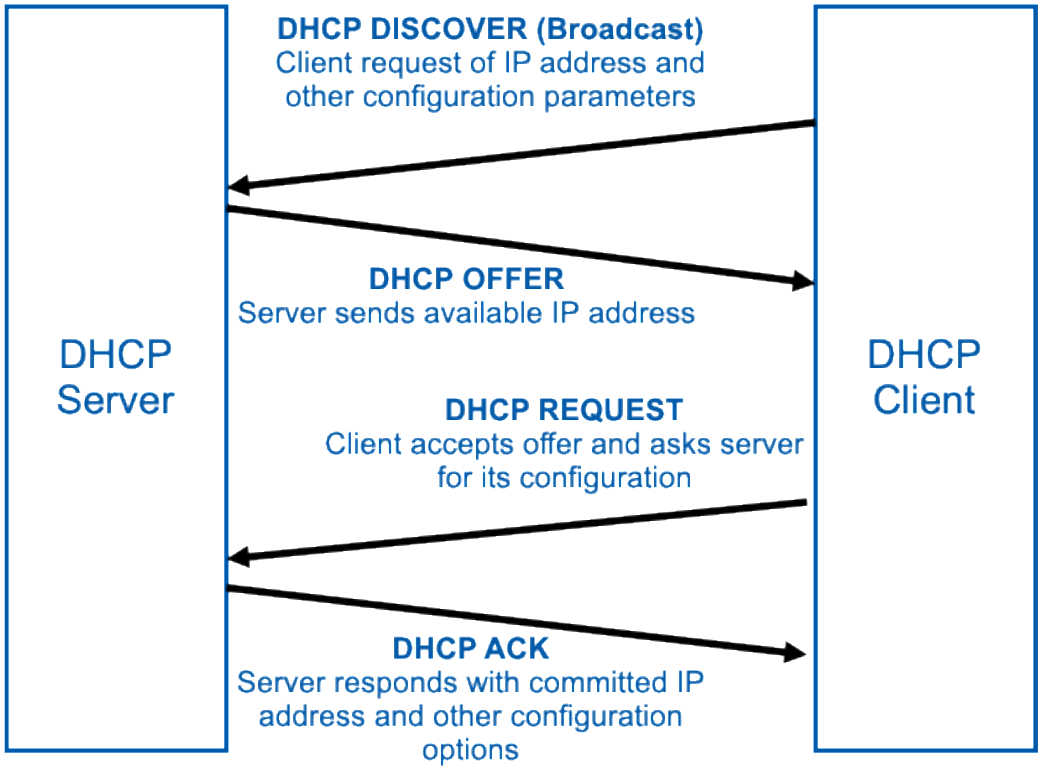
- Simplifies installation and configuration of end systems
- Allows for manual and automatic IP address assignment
- May provide additional configuration information
  - (DNS server, netmask, default router, etc.)

Client broadcasts DHCP DISCOVER packet

- Server answers

Address is assigned for limited time only

- Before the 'lease' expires, client must renew it
- Allows to reclaim addresses of disappearing hosts



DHCP Configuration

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Configuration file:

- Global parameters
  - define global parameters like lease time and domain name
  - parameters are also valid for all subsections

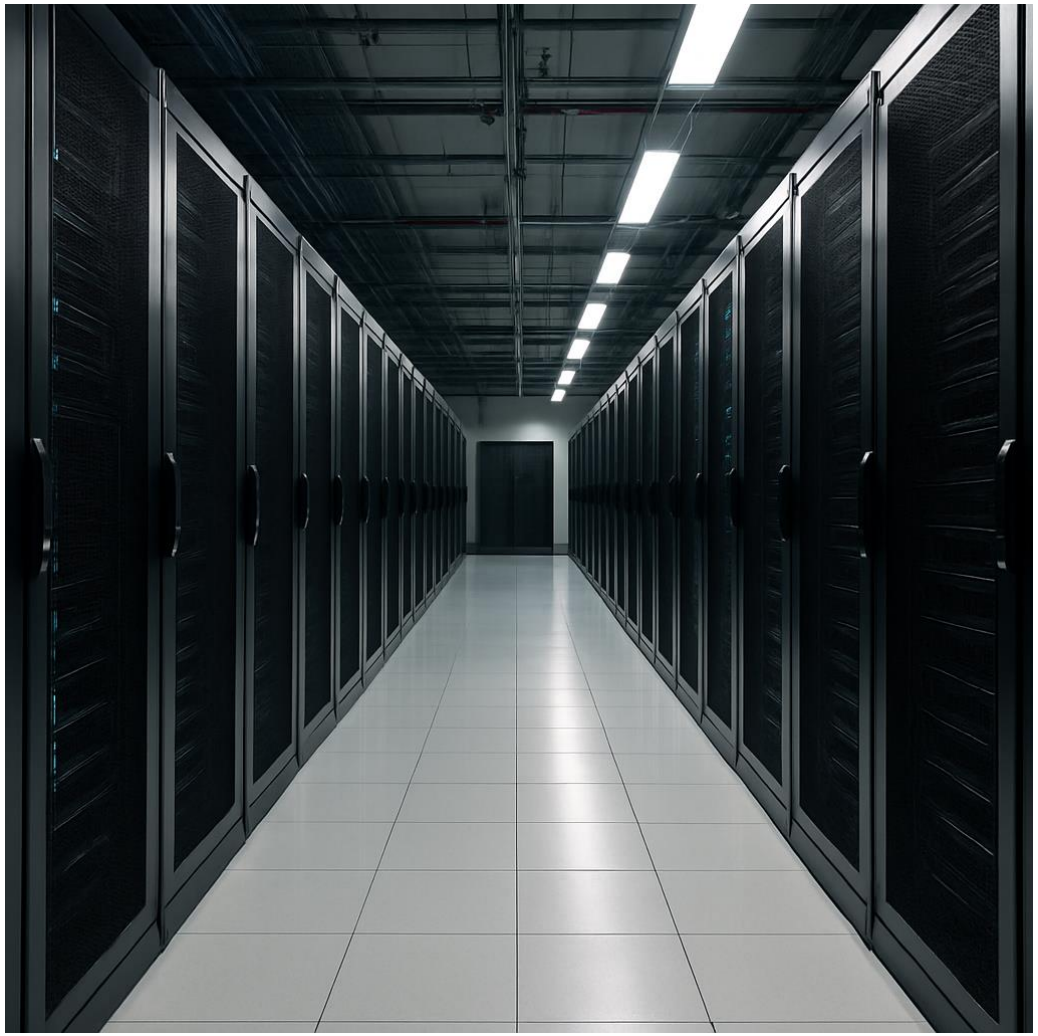
e.g.,

- default time is 24 hours

```
max-lease-time 3600;
default-lease-time 3600;

option domain-name "acme.com";
option domain-name-servers  a.b.c.d, a.b.c.e;
option netbios-name-servers a.b.c.f, a.b.c.g;
option netbios-node-type 8;
```

How do you get to a datacenter now?





How to find routes on Layer 3 (in the Internet)?

The Network Layer

Transport segment from sending to receiving host

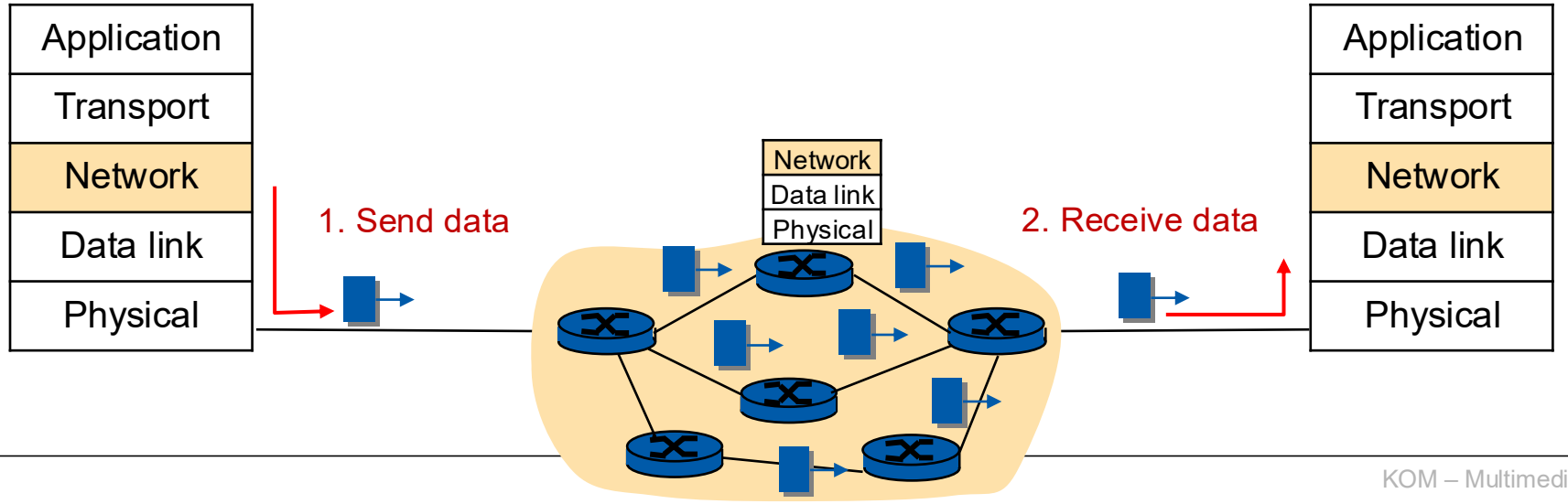
Network layer protocols in every host and router

Routers

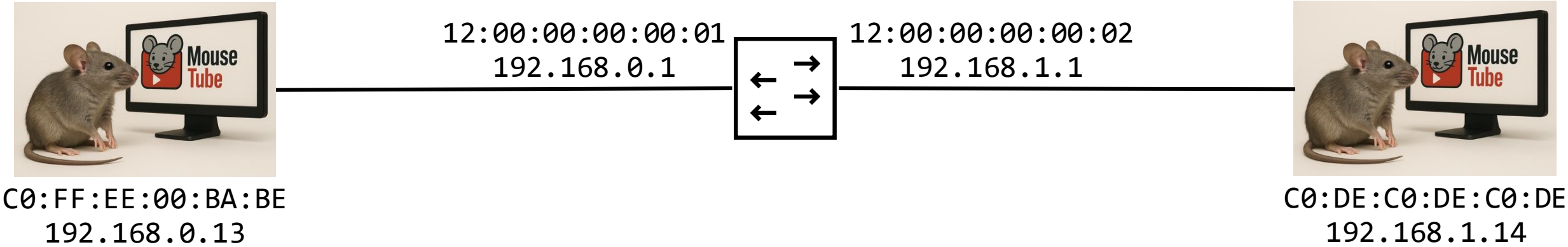
- No state about end-to-end connections
- No network-level concept of “connection”

Packets forwarded using destination host address

- Packets between same source-destination pair may take different paths



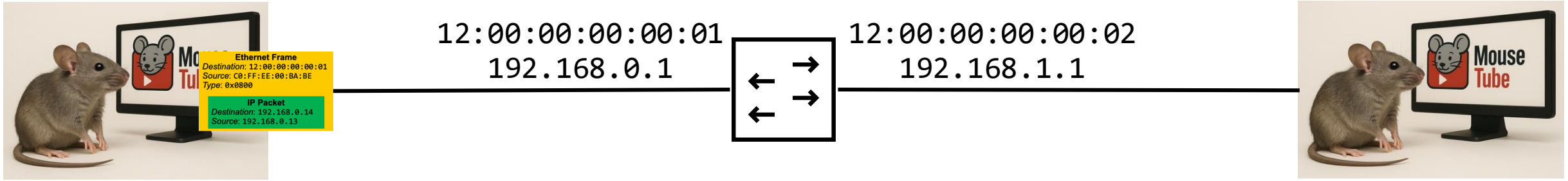
IP
Packet transmission



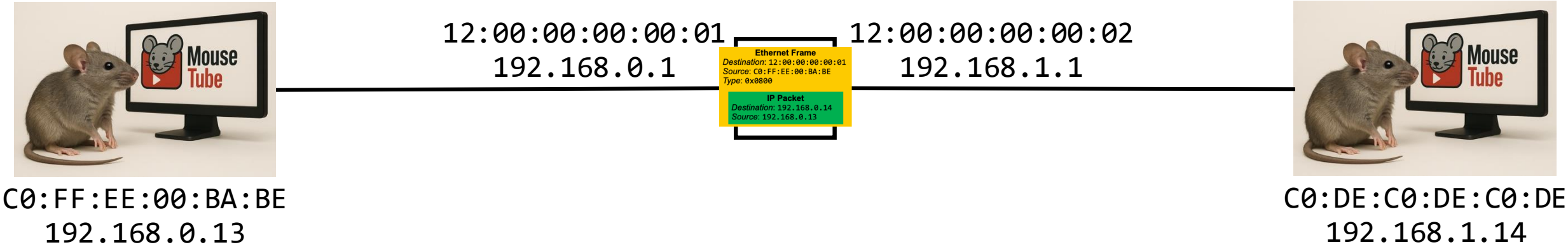
Ethernet Frame
Destination: 12:00:00:00:00:01
Source: C0:FF:EE:00:BA:BE
Type: 0x0800

IP Packet
Destination: 192.168.0.14
Source: 192.168.0.13

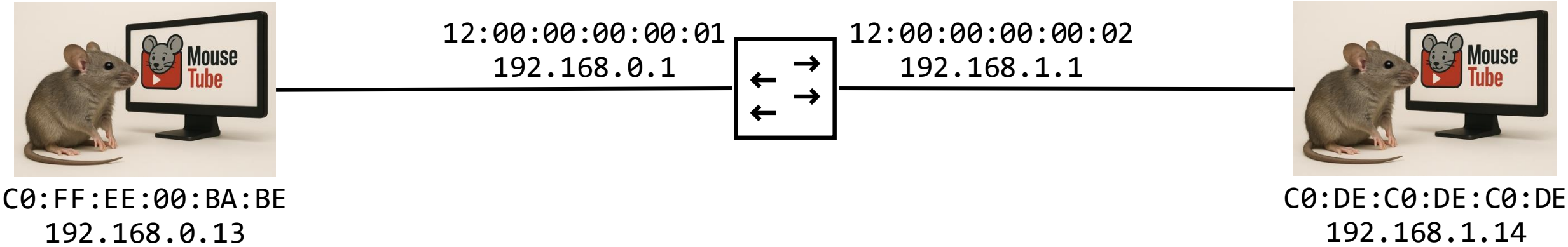
IP
Packet transmission



IP
Packet transmission



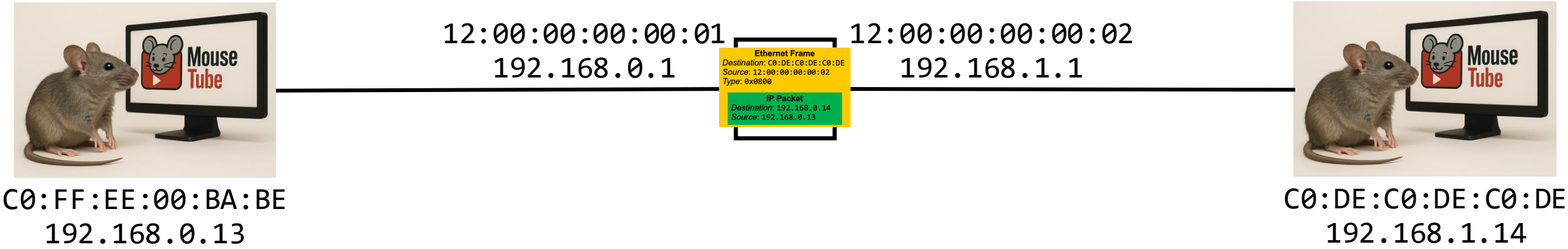
IP
Packet transmission



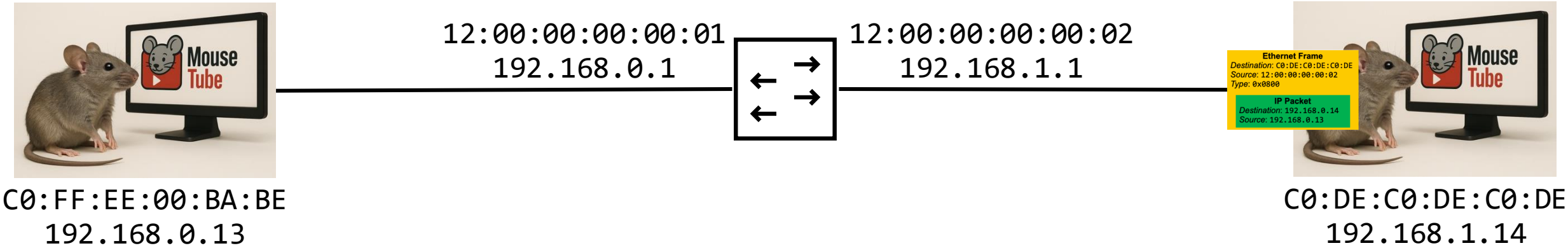
Ethernet Frame
Destination: C0:DE:C0:DE:C0:DE
Source: 12:00:00:00:00:02
Type: 0x0800

IP Packet
Destination: 192.168.0.14
Source: 192.168.0.13

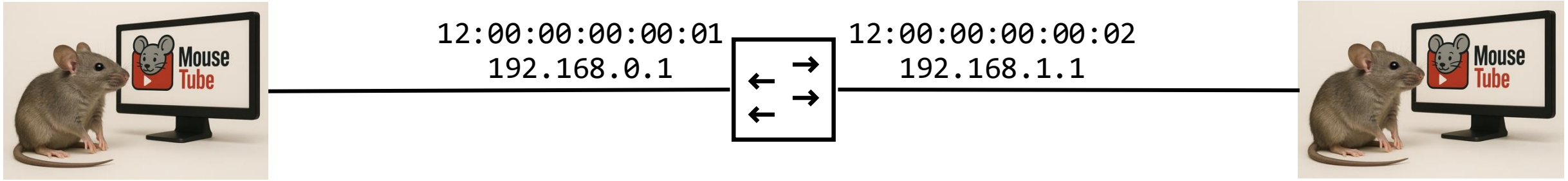
IP
Packet transmission



IP
Packet transmission



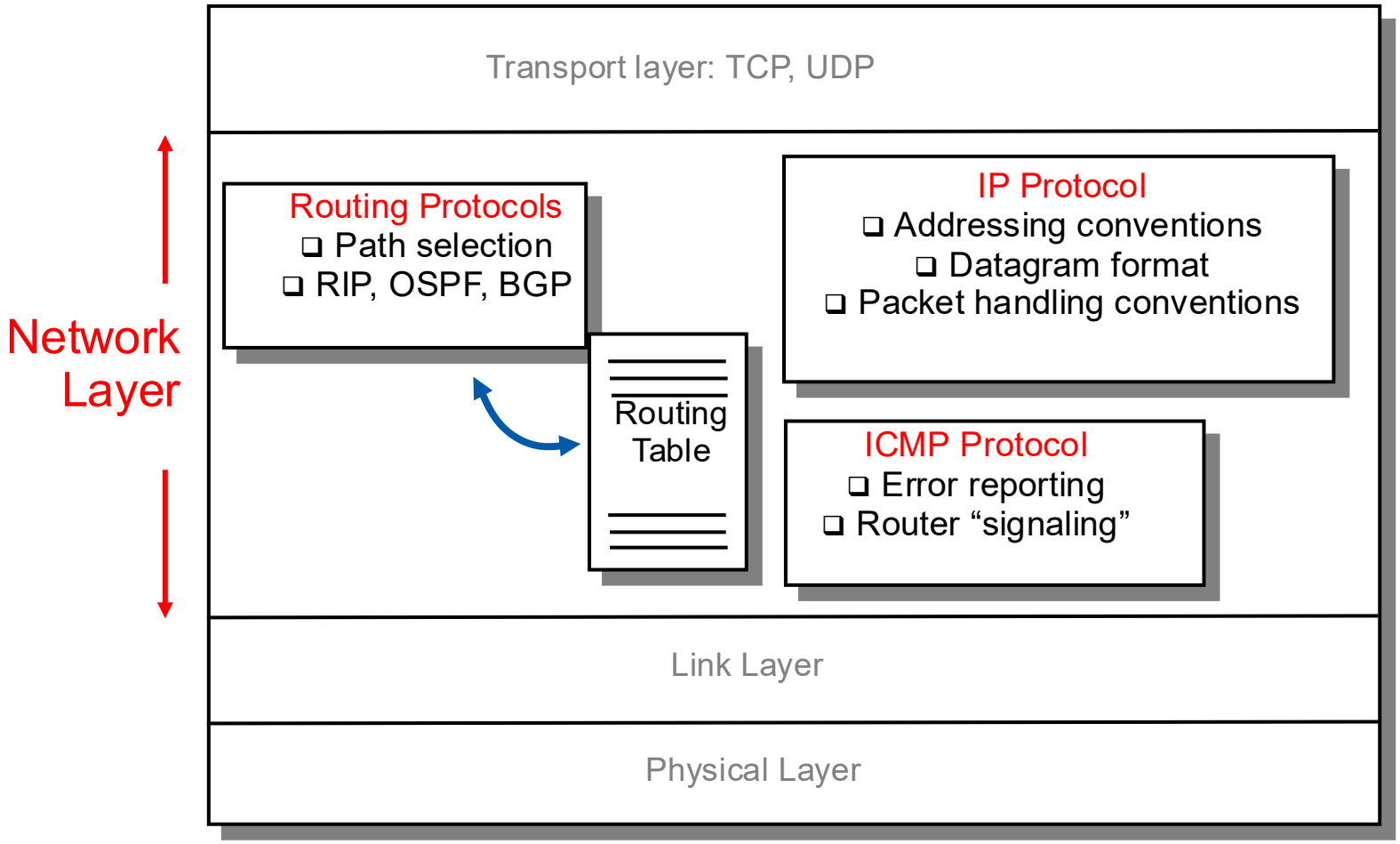
IP
Packet transmission



Ethernet Frame
Destination: C0:DE:C0:DE:C0:DE
Source: 12:00:00:00:00:02
Type: 0x0800

IP Packet
Destination: 192.168.0.14
Source: 192.168.0.13

The Network Layer

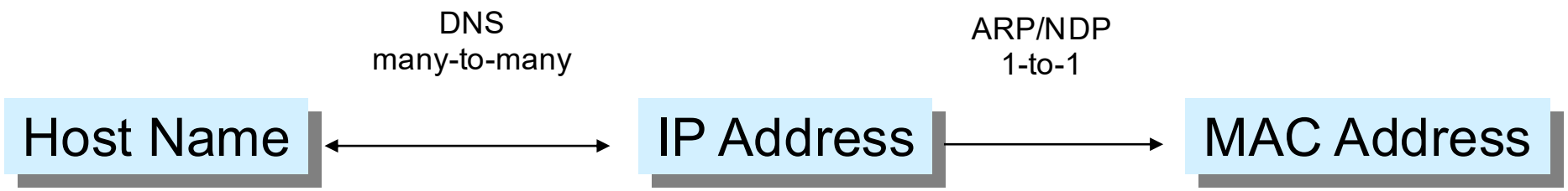


Internet Names and Addresses

	Example	Organization
MAC address	`C0:FF:EE:00:BA:BE`	flat, permanent

IPv4/IPv6 address	132.151.1.35
	`f00d:f00d:f00d:f00d:`
	`f00d:f00d:f00d:f00d`
		topological (mostly)

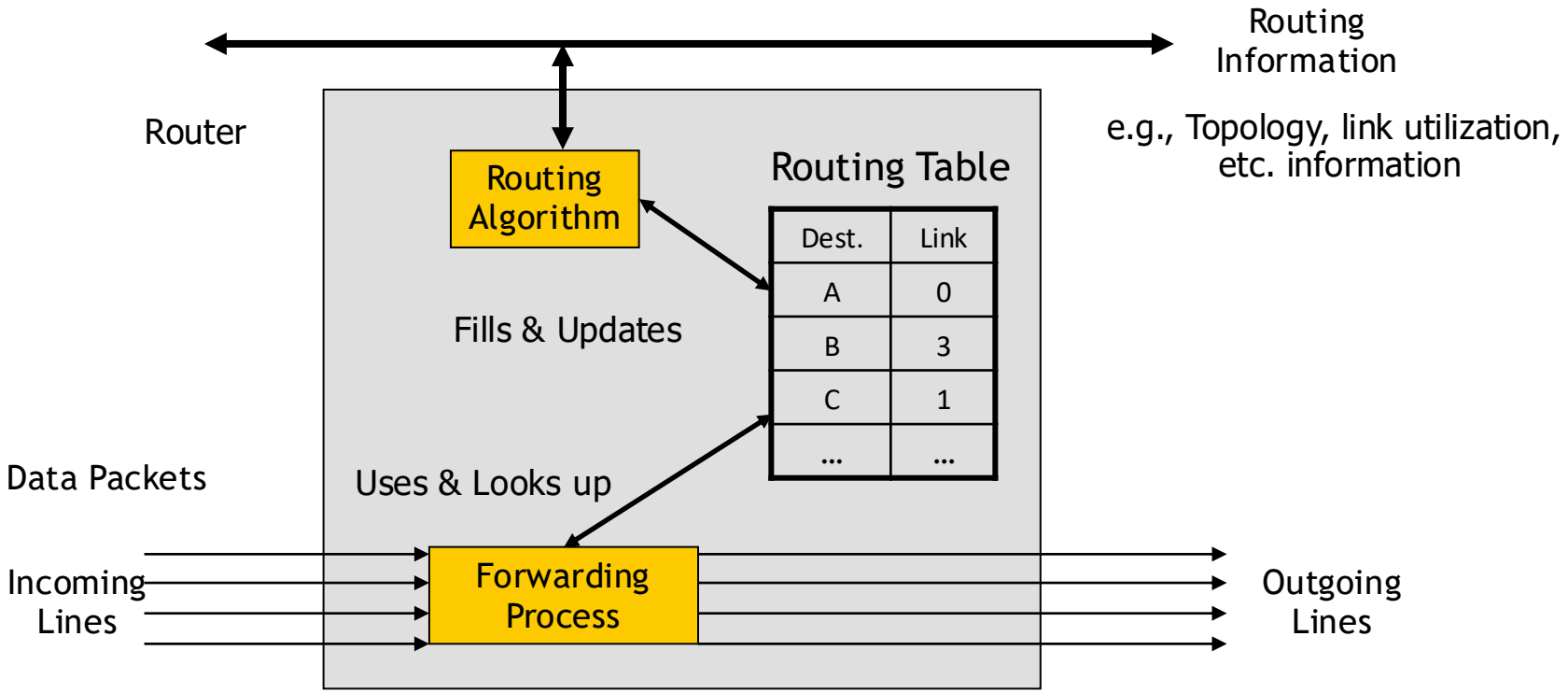
Host name	www.ietf.org	hierarchical



What Happens in the Routers?

Intermediate Systems (routers) on the Internet have two main functions

- Routing: Determine which route to use
- Forwarding: What happens when a packet arrives?



IPv4 Addressing

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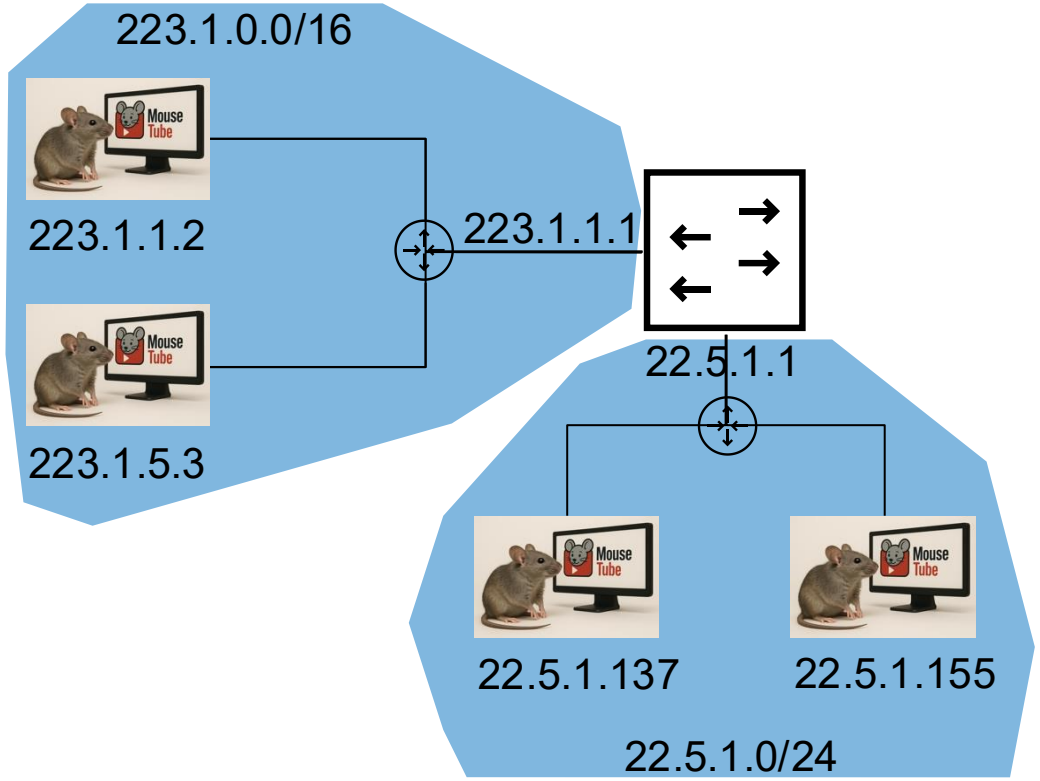
IPv4 address: 32-bit identifier for host or router interface

223.1.1.1 = 11011111 00000001 00000001 00000001

223 1 1 1

Interface: connection between host/router and physical link

- Router's typically have multiple interfaces
- Host may have multiple interfaces
- IP addresses associated with each interface

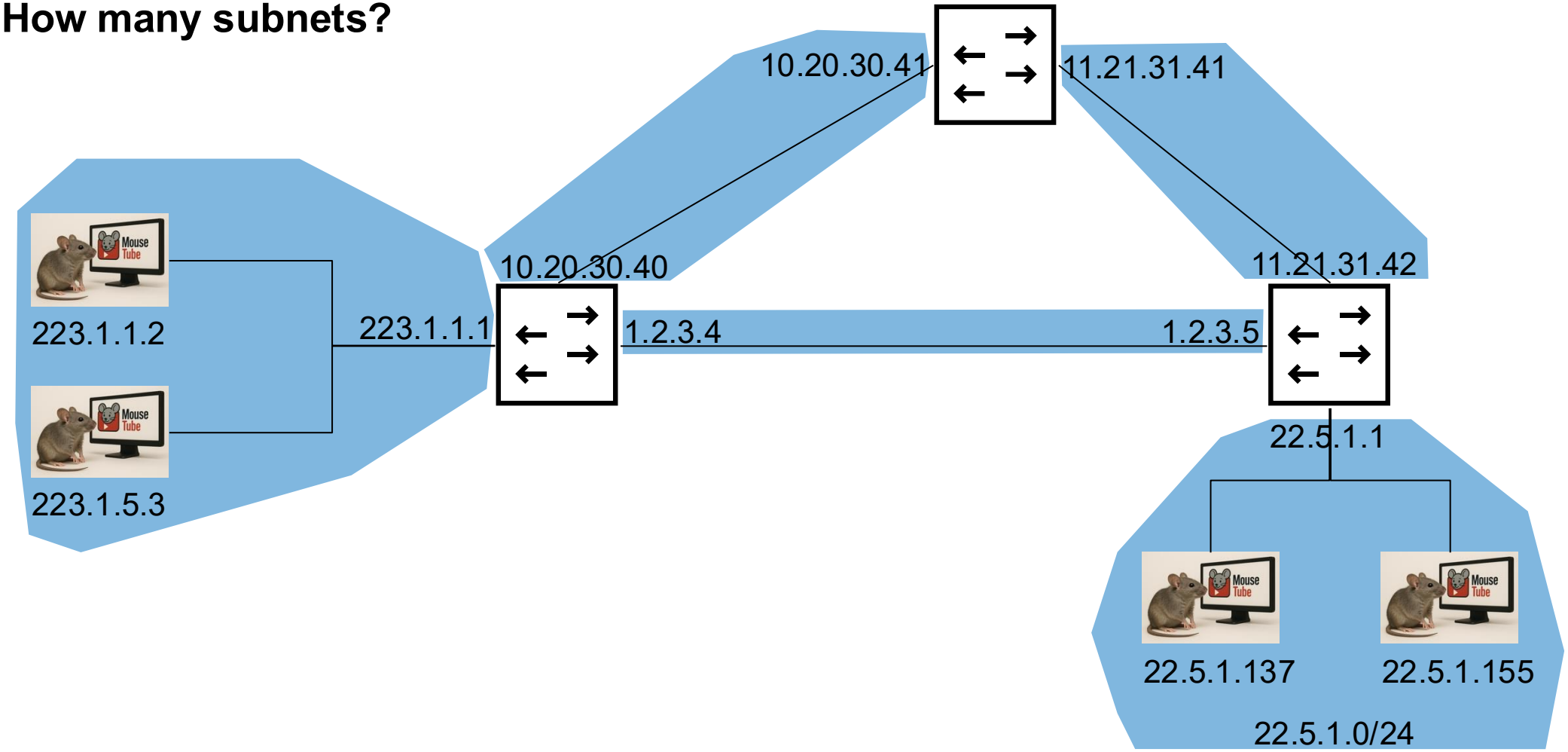


What's a subnet?

- Device interfaces with same network part of IP address
- Can physically reach each other without intervening router

Subnets

How many subnets?

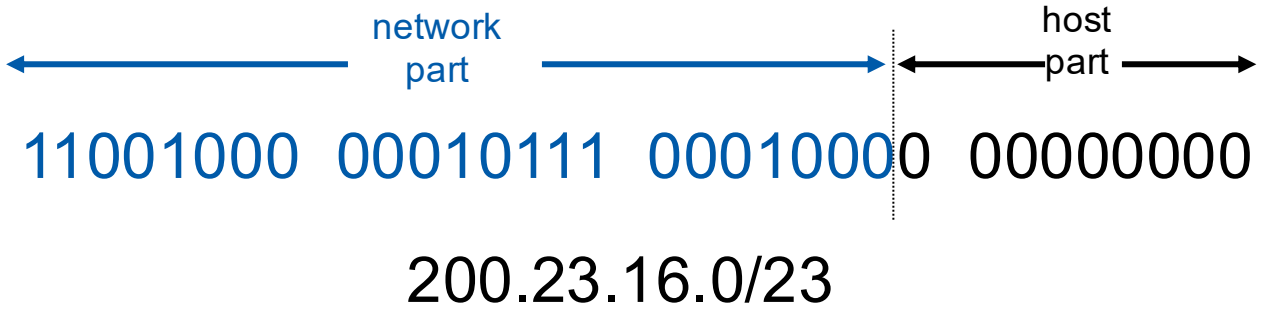


Network Part of IP Address - CIDR

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CIDR: Classless InterDomain Routing

- Network portion of address of arbitrary length
- Address format: a.b.c.d/x, where x is # bits (the network mask) in network portion of address



Routers store only network address/subnet not address of single hosts

- Example: All sites in Europe common prefix
  → Only single entry in most U.S. routers
- "Longest prefix match routing" on masked address, e.g., 200.23.16.0/22

Address/mask	Next hop
200.23.16.0/22	1
200.23.16.0/23	2
200.23.10.0/17	3

Network Part of IP Address - CIDR

CIDR: Classless InterDomain Routing

- Network portion of address of arbitrary length
- Address format: a.b.c.d/x, where x is # bits (the network mask) in network portion of address




```
ubuntu@ubuntu:~$ ip a
2: ens18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether cd:11:98:fc:2d:32 brd ff:ff:ff:ff:ff:ff
    altname enp0s18
    inet 10.2.2.184/16 metric 100 brd 10.2.255.255 scope global ens18
       valid_lft forever preferred_lft forever
    inet6 2001:41b8:810:41:be24:11ff:feff:1f16/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86260sec preferred_lft 14260sec
    inet6 fe80::be24:11ff:feff:1f16/64 scope link
       valid_lft forever preferred_lft forever
```

Routers st

- Example: All sites in Europe common prefix
  → Only single entry in most U.S. routers
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Address/mask	Next hop
200.23.16.0/22	1
200.23.16.0/23	2
200.23.10.0/17	3

Longest Prefix Matching at Routers

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Routing table of a router

Network	Binary Notation	Link Interface
200.23.16.0/21	11001000 00010111 00010	0
200.23.22.0/24	11001000 00010111 00010110	1
200.23.24.0/16	11001000 00010111 00011000	2
otherwise		3

Example

Packet Destination: 200.23.22.161
11001000 00010111 00010110 10100001

Which interface?

Hierarchical Routing

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Many routing lectures are idealization

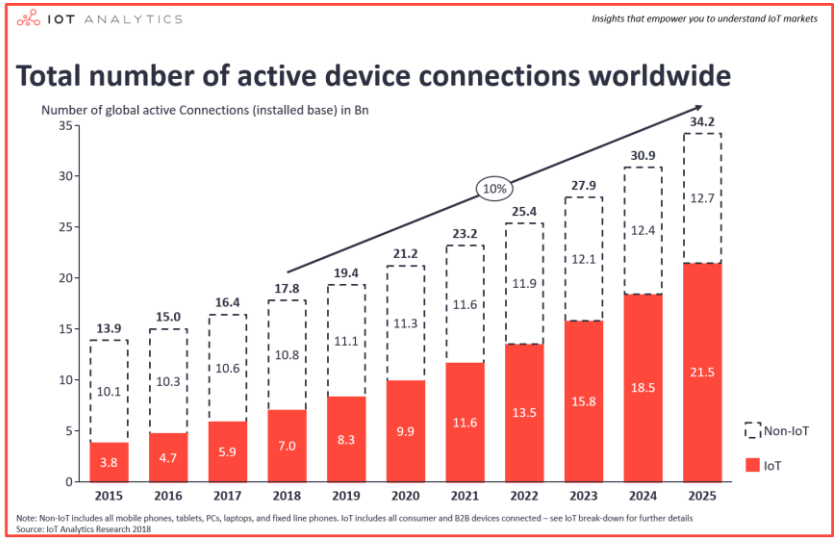
- All routers are assumed to be identical
- Network assumed to be “flat”
  … Real world, however, looks different

Scale (~19 billion destinations!):

- Can’t store all destinations in routing tables
- Routing table exchange would overload links

Administrative autonomy:

- Internet = network of networks
- Each network admin may want to control routing in its own network



Routing on the Internet

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The Global Internet consists of Autonomous Systems (AS) interconnected with each other:

- Stub AS: small corporation (only one link to the Internet)
- Multihomed AS: large corporation (multiple links, but no transit)
- Transit AS: provider

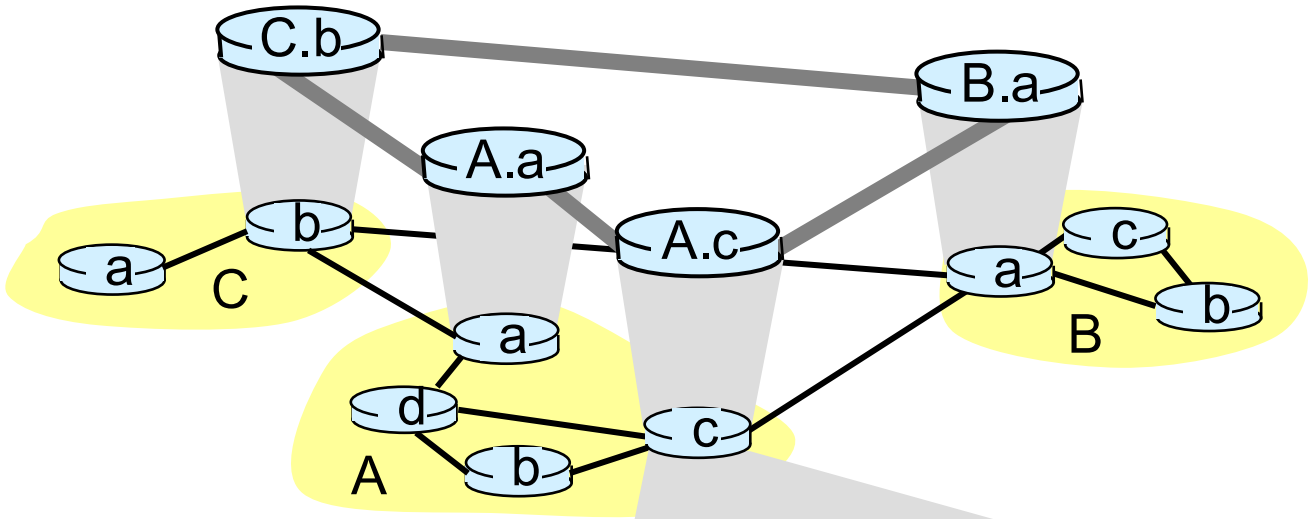
On the Internet today we see many autonomous systems (>100k in 2021)

- Have different sizes
- Exchange services with each other as equals or as provider/customer
- Have different relations to each other

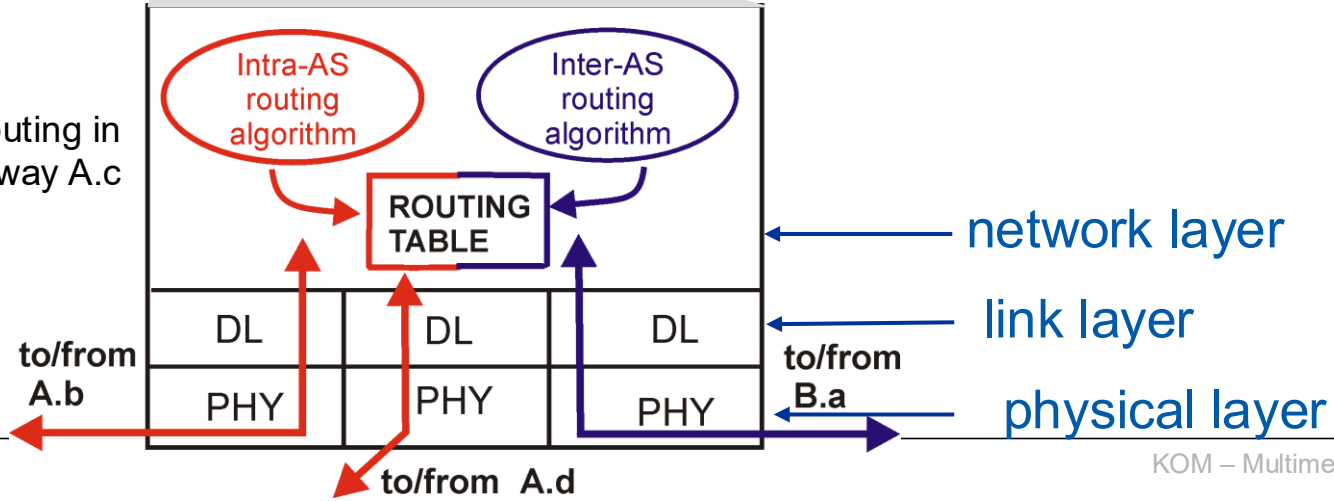
Every AS has a unique number (32bit, formerly 16 bit)

Every AS must know a route to every network

Autonomous Systems and Hierarchical Routing



inter-AS, intra-AS routing in gateway A.c



- Aggregate routers into regions: "Autonomous Systems" (AS)
- Gateway routers
  - Inter-AS routing amongst themselves
  - Intra-AS routers with other routers in their AS
  - responsible for routing to destinations outside AS

Intra/Inter-AS routing

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Two-level routing:

Intra-AS: administrator is responsible for choice

- Routing Information Protocol (RIP): Distance Vector
- Open Shortest Path First (OSPF): Link State
- Interior Gateway Routing Protocol (IGRP): Distance Vector (Cisco proprietary)
- Intermediate System to Intermediate System (IS-IS): Link-State protocol

Inter-AS: unique standard

- Border Gateway Protocol (BGP): Path Vector
  (sort of distance vector, but with path information for loop avoidance)

Why Different Inter- and Intra-Domain Routing?

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Scale:

- Hierarchical routing saves table size, reduced update traffic

Policy:

- Intra-AS: single admin, so no policy decisions needed
- Inter-AS: admin wants control over how its traffic routed, who routes through its network

Performance:

- Intra-AS: can focus on performance
- Inter-AS: policy may dominate over performance

What happens when you connect your laptop to a network?

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We know

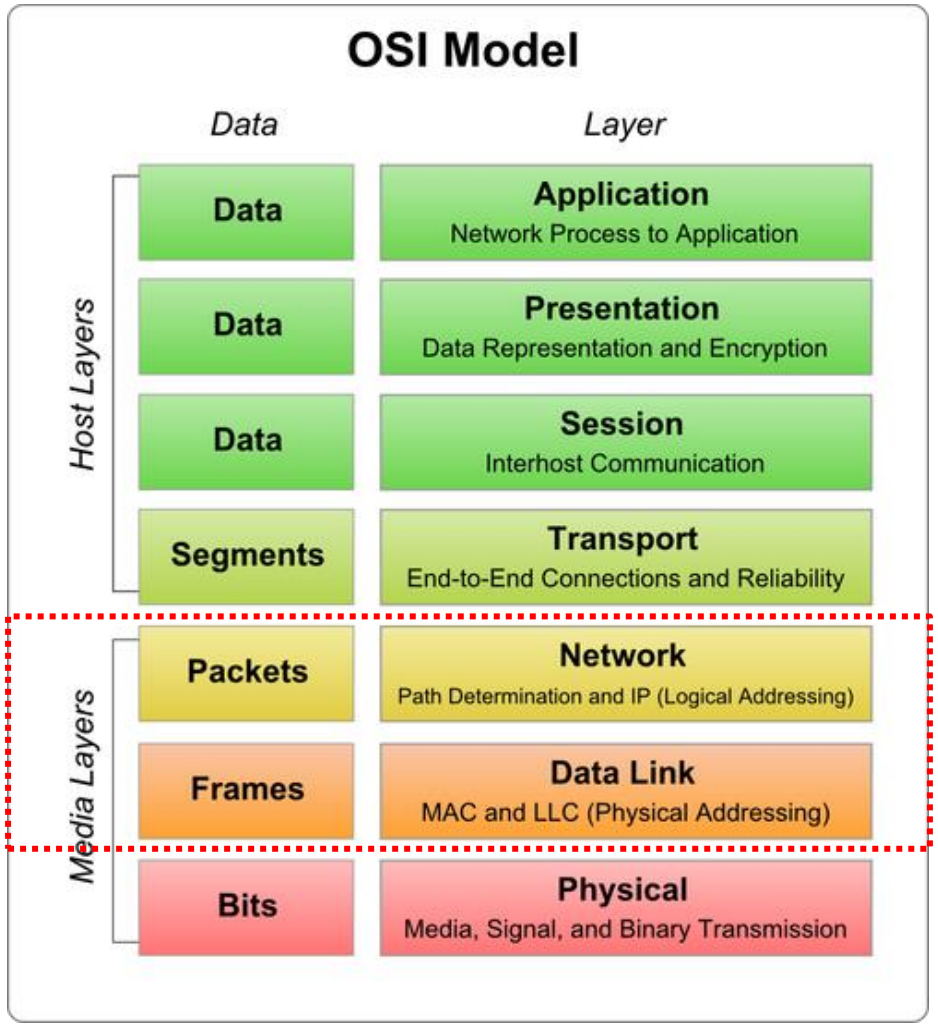
- How frames are forwarded on Layer 2
- How packets are routed on Layer 3

How are those two connected?

- Address Resolution Protocol (ARP) for IPv4
- Neighbor Discovery Protocol (NDP) for IPv6

When your computer sends a packet towards a server in the Internet

- It sends the packet to the router first
- Who then forwards it further via your ISPs network



How does your browser/application send/receive data?

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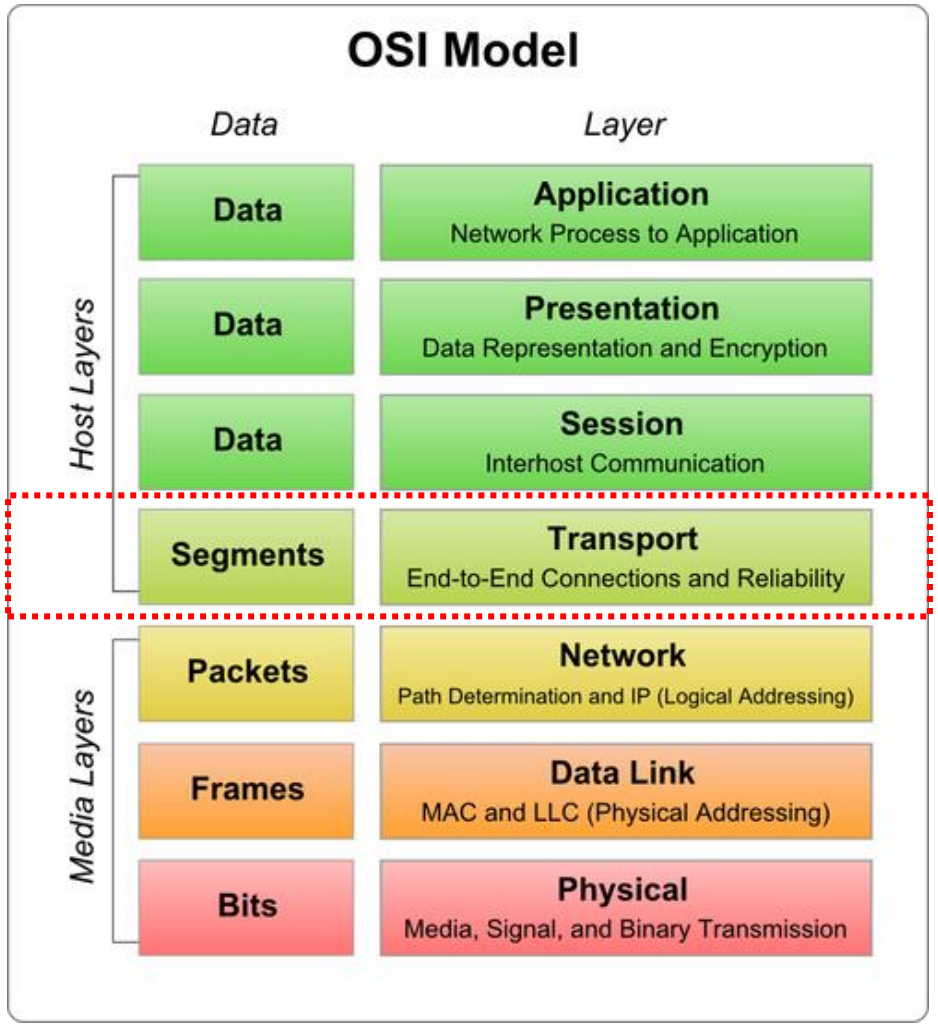
Inter-process end-to-end communication via communication networks

- For example, a process on your machine (the browser) communicates with a webserver

Transport layer protocols in the Internet

- UDP = User Datagram Protocol
- TCP = Transmission Control Protocol
- …

Network layer allows for communication between hosts, transport layer for communication between applications



Ports: Addressing Concept

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Address identifies

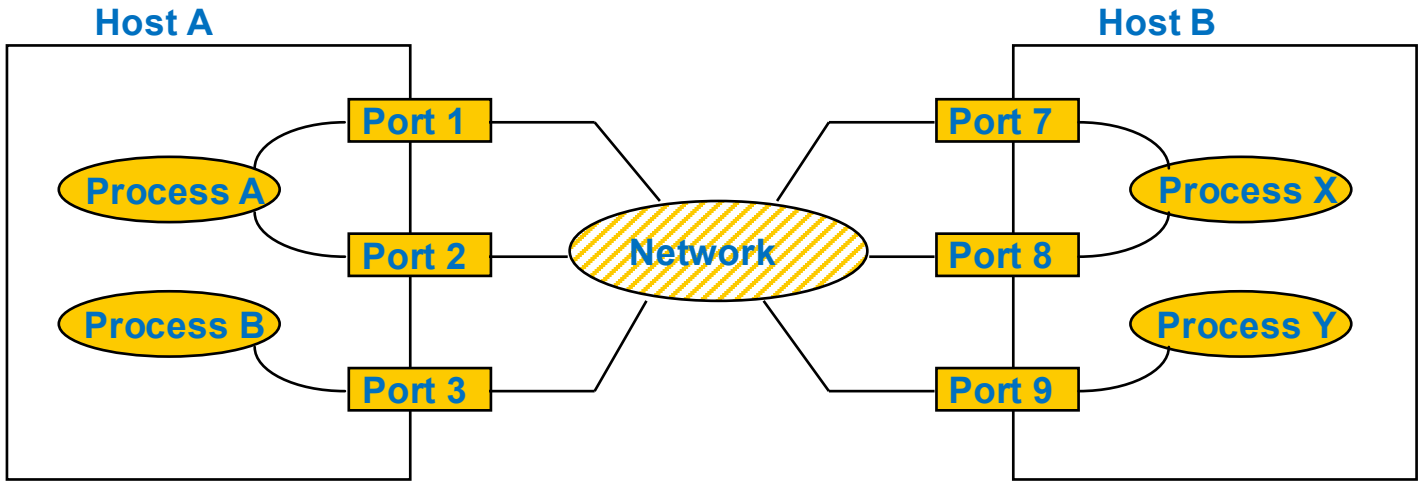
- type of service or application (destination of communication)

In order to communicate, the sender needs to know

- Network layer (IP) address of receiving host
- Port number of receiving service
  - and the used transport protocol

The same process may supply multiple ports

- E.g., webservers can provide 80 and 443

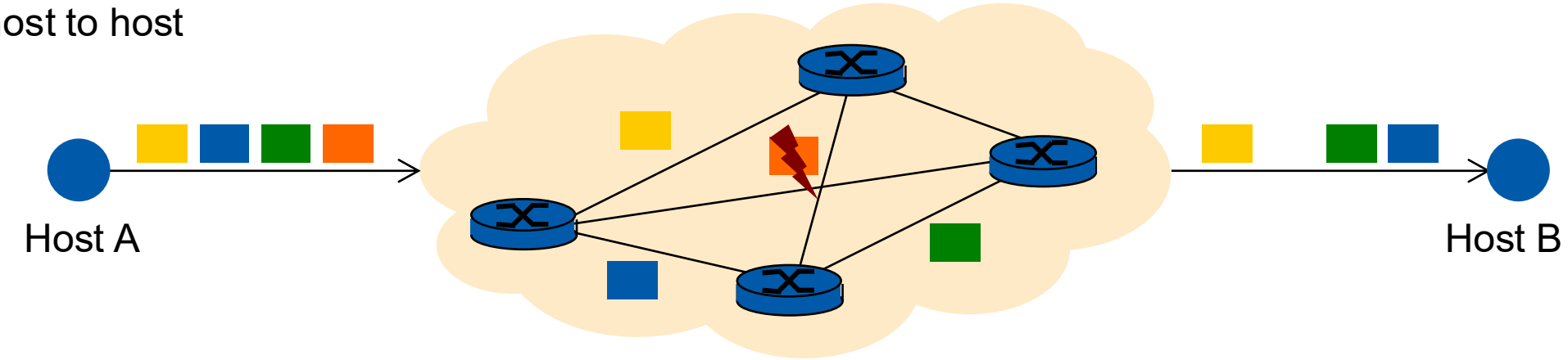


UDP – User Datagram Protocol

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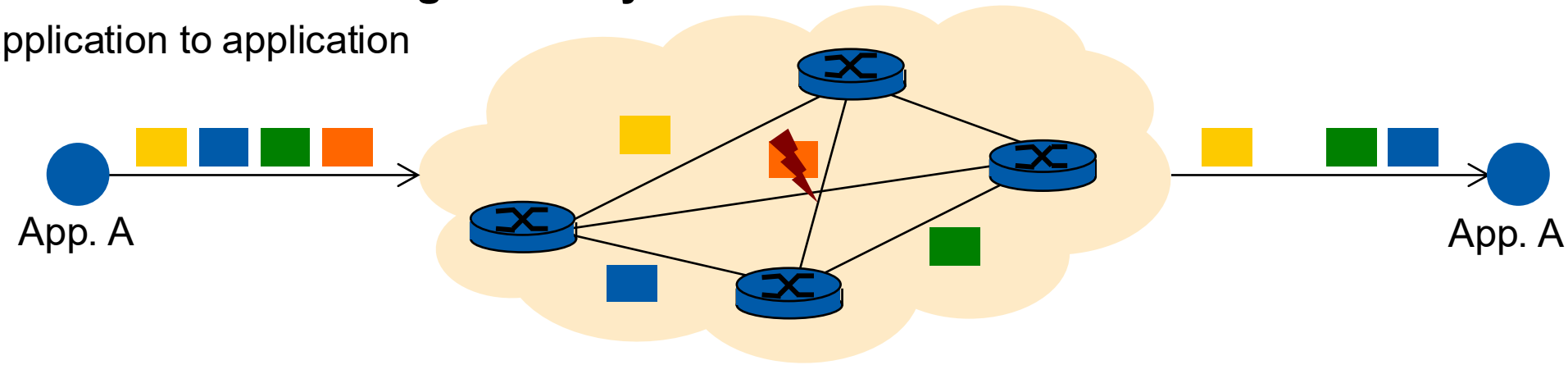
Internet layer offers best effort packet delivery

- From host to host



UDP offers best effort message delivery

- From application to application



UDP – User Datagram Protocol - RECAP

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Benefits of UDP

- needs very few resources
  - in end systems, in each data unit
- no connection establishment
  - (hence, no overhead, faster transmission)
- simple implementation possible
  - (e.g. suitable for boot PROM)
- applications can precisely control
  - packet flow
  - error handling/reliability
  - Timing

Suitable for simple client-server interactions, i.e. typically

- 1 request packet from client to server
- 1 response packet from server to client

Used by e.g.

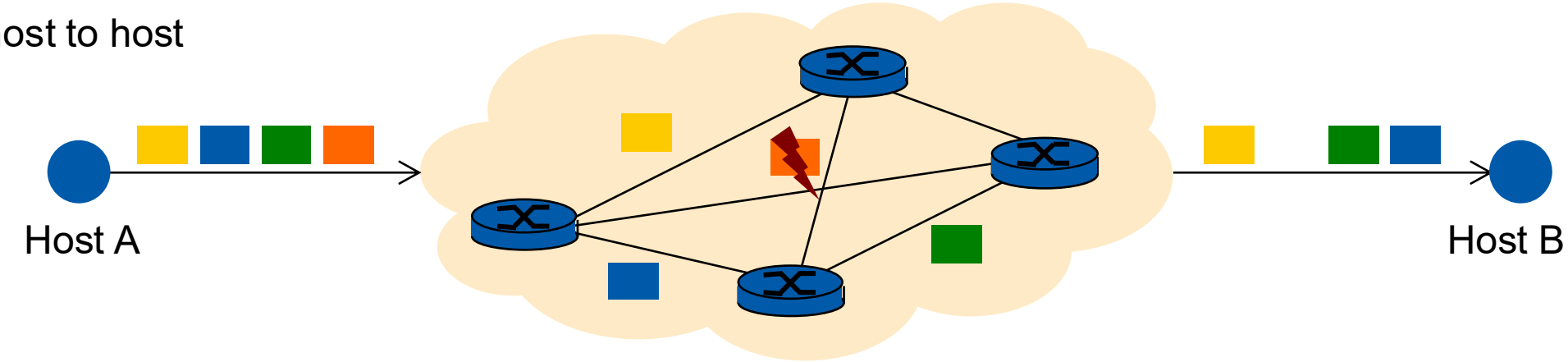
- DNS: Domain Name Service
- SNMP: Simple Network Management Protocol
- NFS: Network File System
- RTP: Real-time Transport Protocol

TCP – Transmission Control Protocol - Basics

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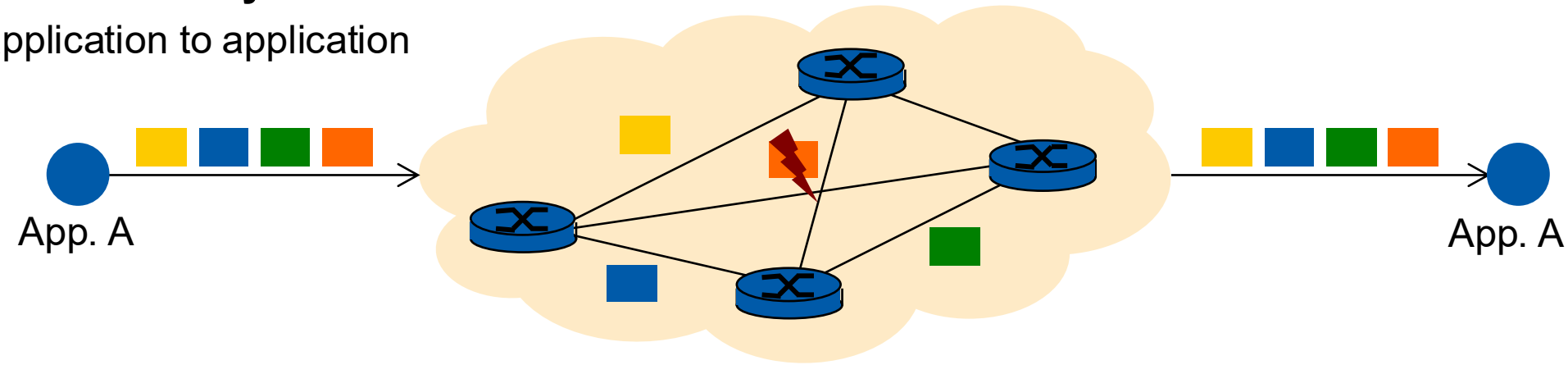
Internet layer offers best effort packet delivery

- From host to host



TCP offers reliable byte stream

- From application to application



Basics of TCP

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Connection-oriented

- before sending data, TCP establishes a connection between sender and receiver

Reliable delivery

- lost data is detected and retransmitted

Ordered delivery

- data arrives to the application in the same order it was sent

Error detection

- checksums help detect corruption

Flow control

- the receiver can limit how much data the sender transmits, preventing buffer overflow

Congestion control

- the sender adjusts its rate when the network appears congested, helping protect overall network stability

How does TCP work?

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Sequence numbers

- track bytes and preserve order

Acknowledgments (ACKs)

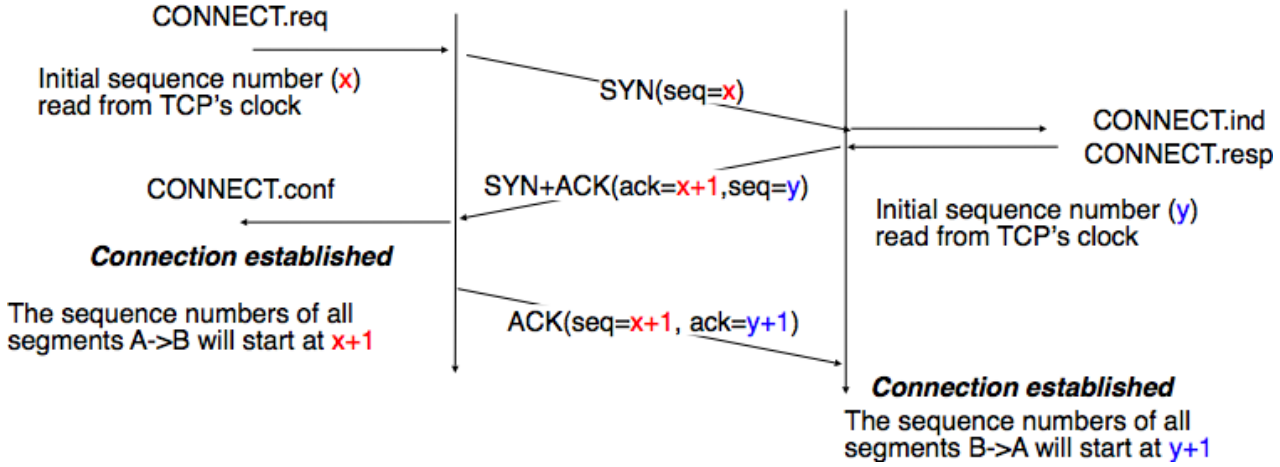
- confirm successful reception

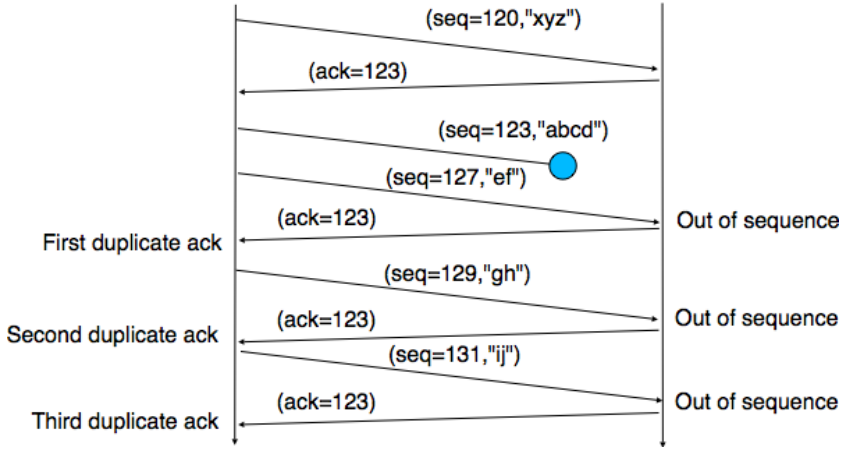
Retransmissions

- resend missing data

Congestion window

- reduces sending rate under congestion, increases it when conditions improve





TCP in Use & Characteristics of TCP

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TCP transport entity at sending side

- Accepts user data streams for local processes
- Splits them into pieces <= 64 KB, typically 1460 bytes to fit into single Ethernet frame with IP and TCP headers)
- Sends each piece as separate IP datagram

TCP transport entity at receiving side

- Gets TCP data from datagram received at host
- Reconstructs original byte streams

Some Missing Characteristics:

no broadcast

- no possibility to address all applications at the same time with a single message

no multicasting

- group addressing not possible

no QoS parameters

- not suited for different media characteristics

no real-time support

- no correct treatment/communications of audio or video possible
- e.g. no Forward Error Correction (FEC)

Summary of TCP

Benefits of TCP

- reliable data transmission
- efficient data transmission despite complexity
- can be used with LAN and WAN for
  - low data rates
    (e.g. interactive terminal)
  - high data rates
    (e.g. file transfer)

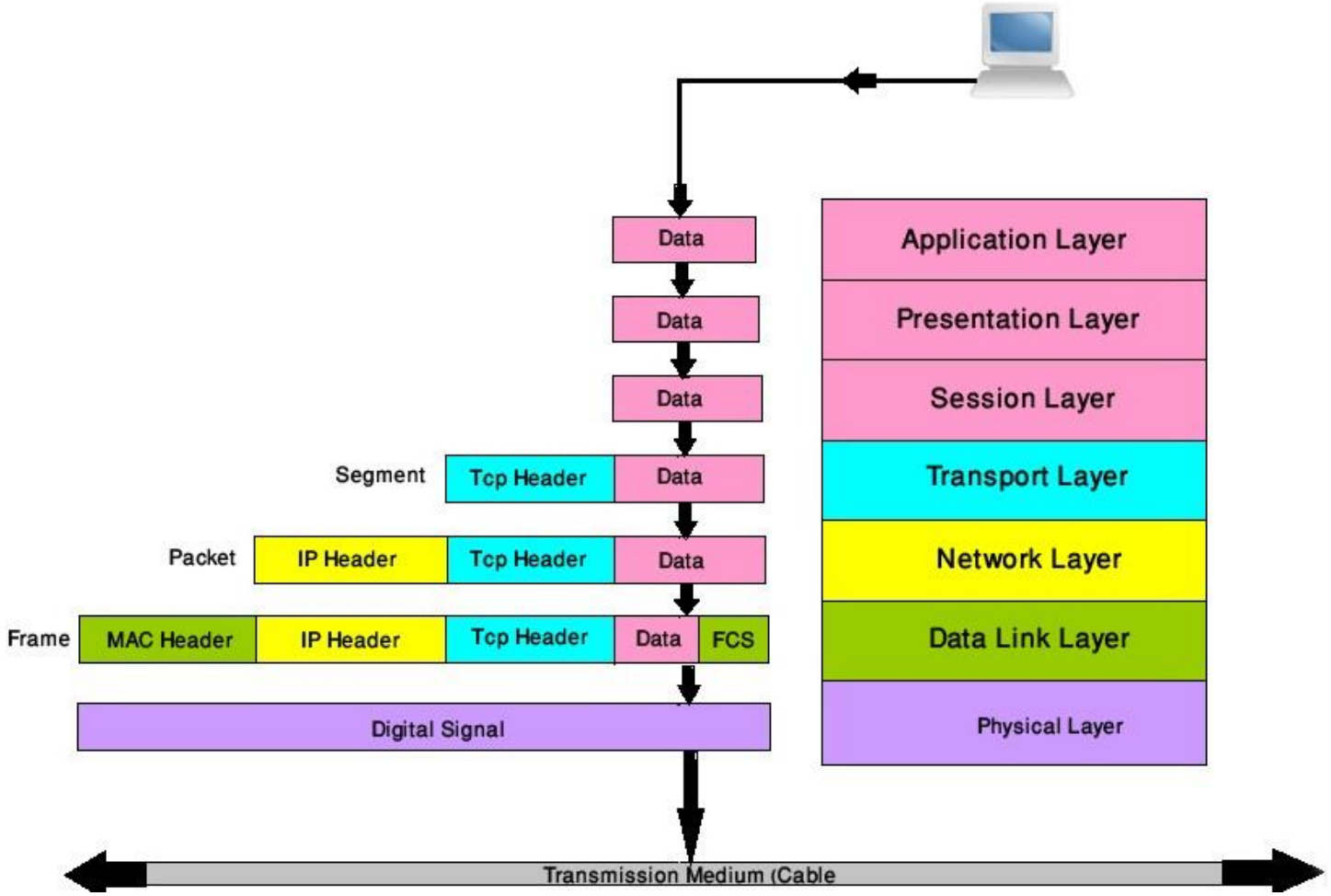
Disadvantages when compared with UDP

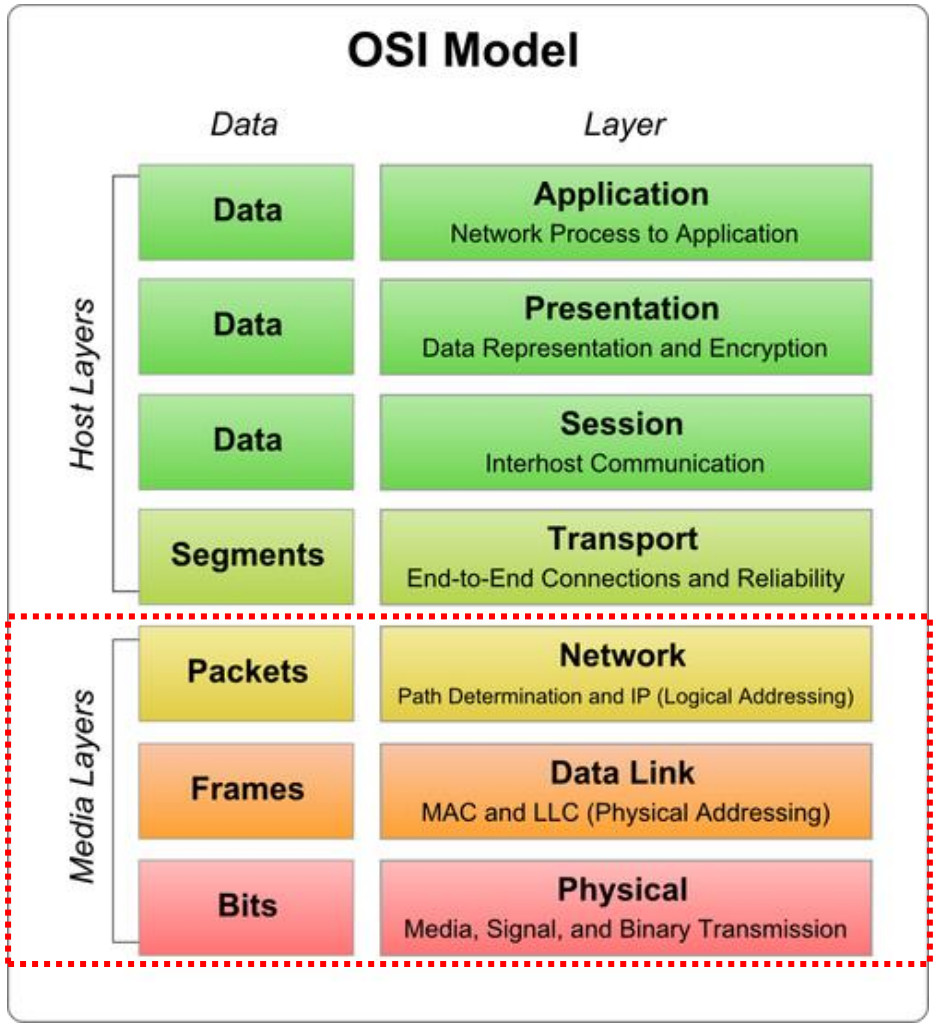
- higher resource requirements
  - buffering, status information, timer usage
  - connection set-up and disconnect necessary
    - even in case of short data transmissions

Applications

- file transfer (FTP)
- e-mail (SMTP)
- Web-Applications via HTTP (at least before QUIC)

What happens when you connect your laptop to a network?



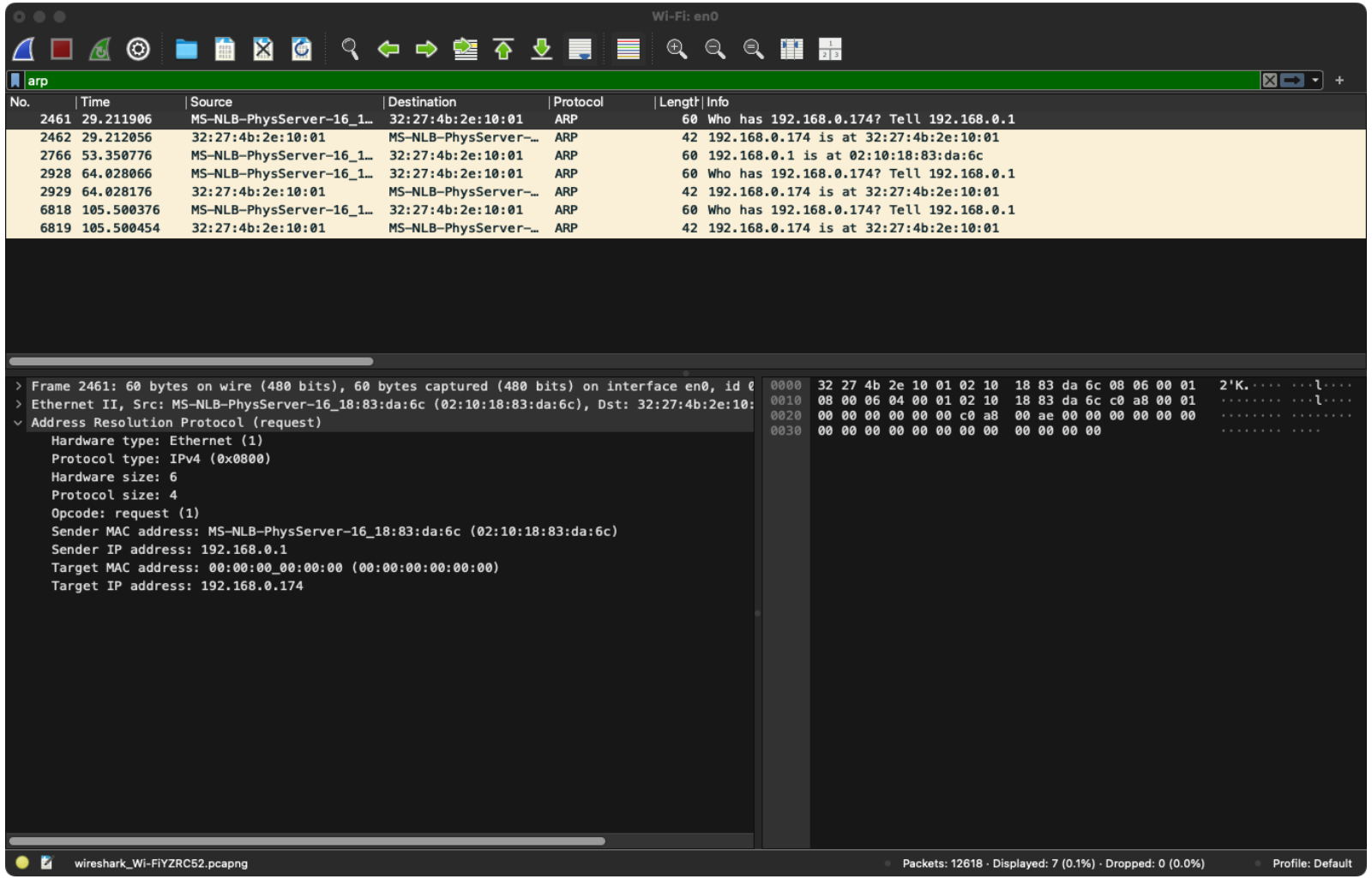


https://blog.smartbuildingsacademy.com/what-is-the-osi-model

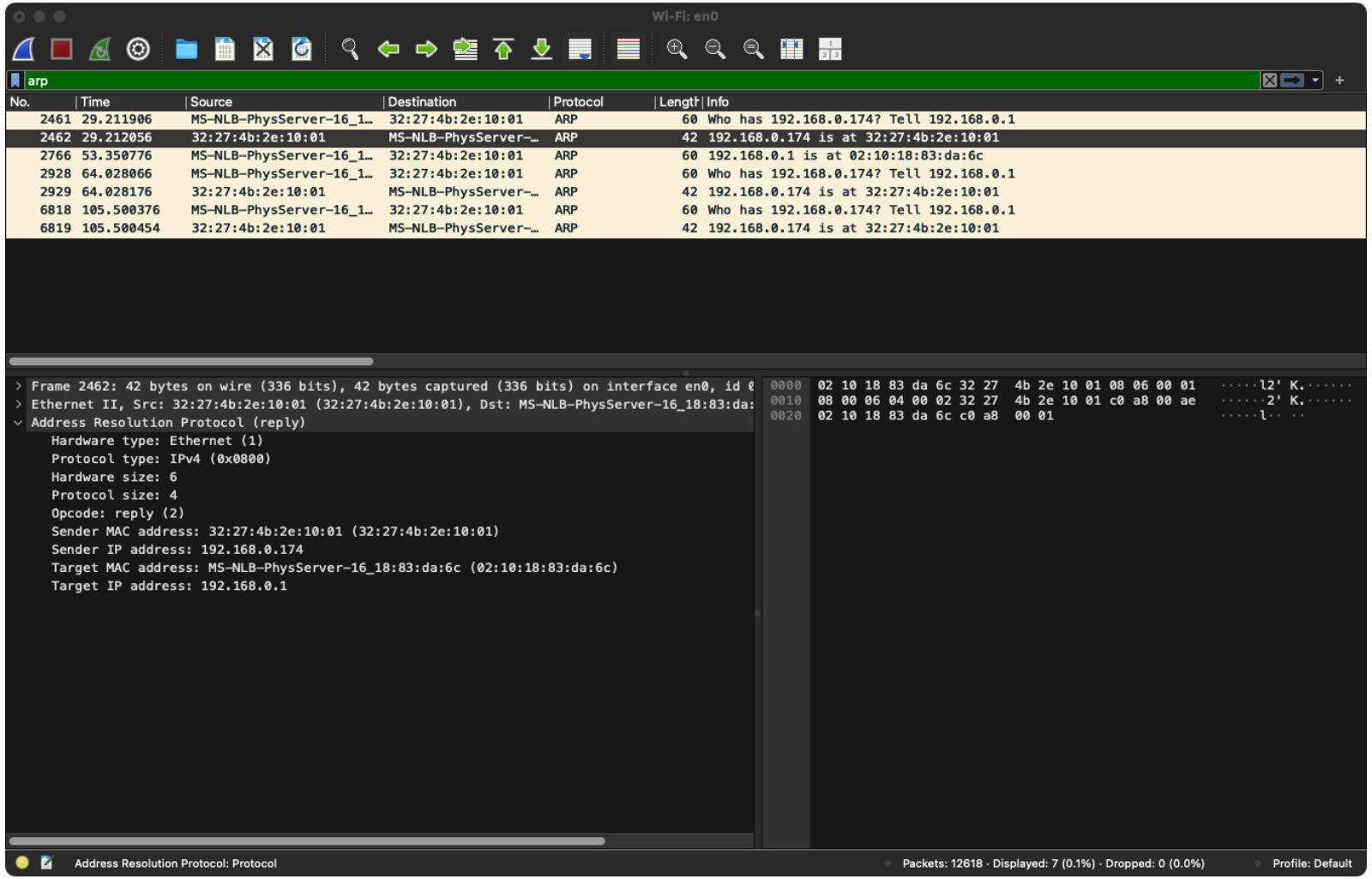
And how does it look like?

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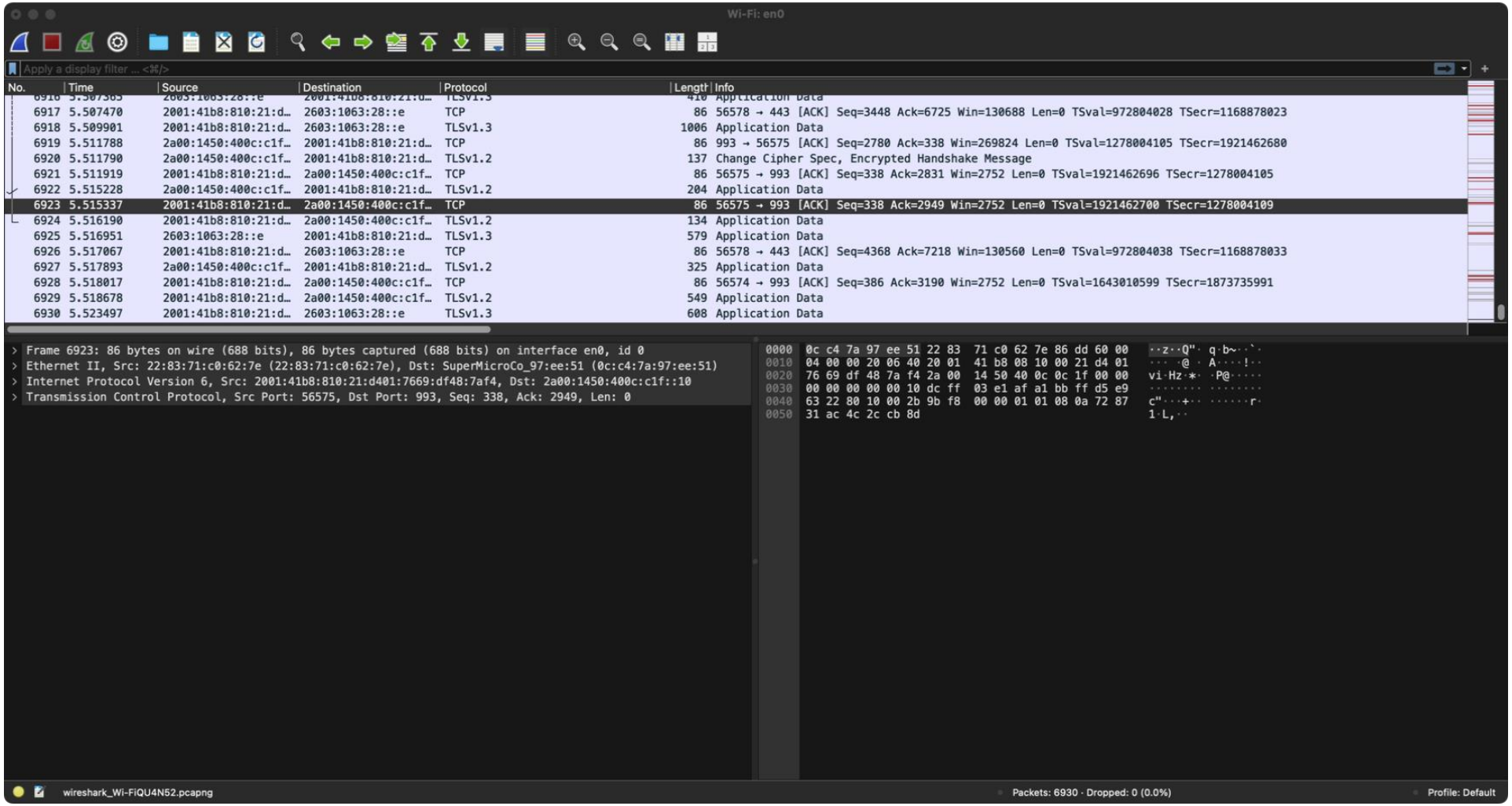
And how does it look like?
ARP Request



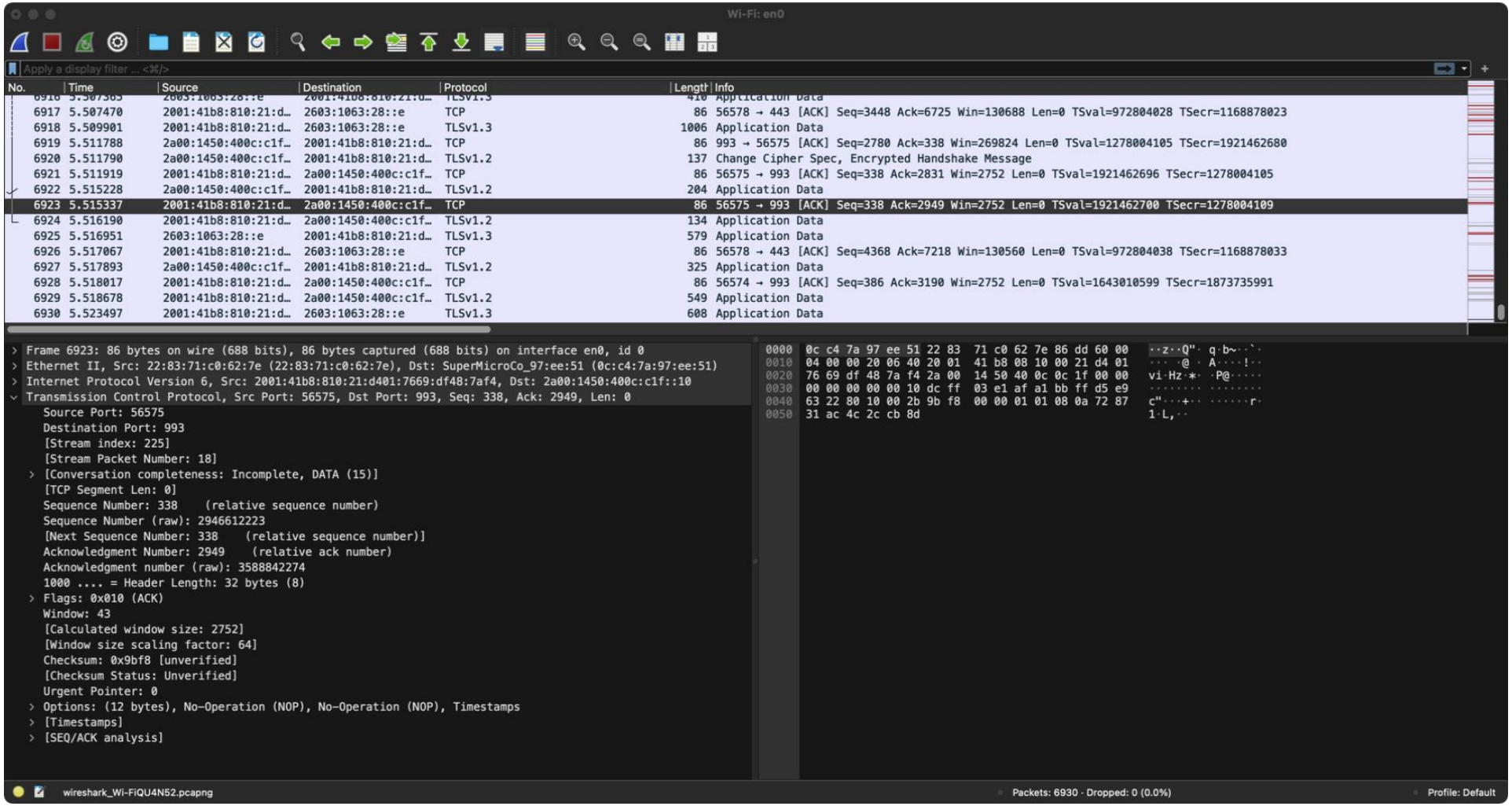
And how does it look like?
ARP Reply



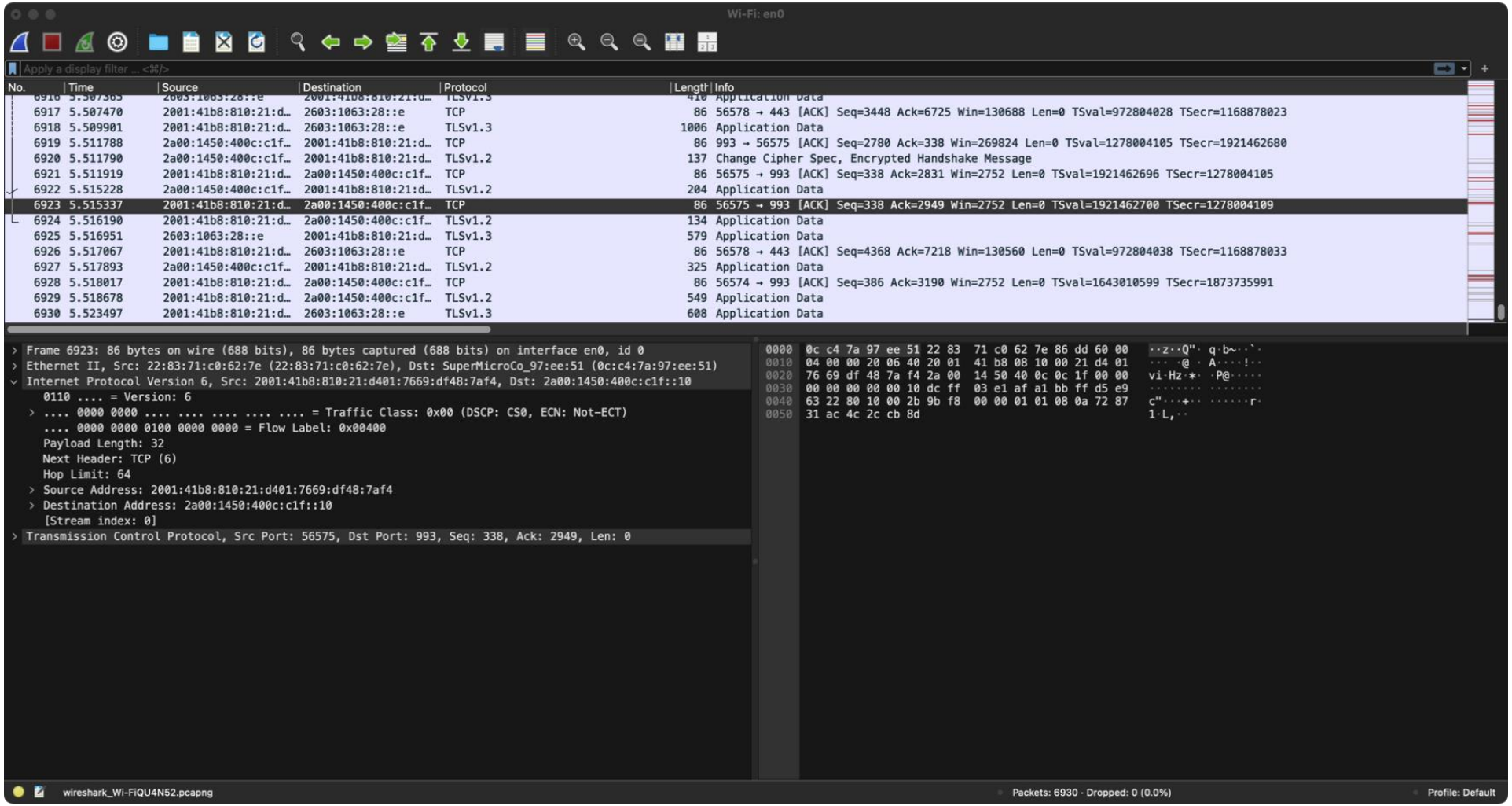
And how does it look like?



And how does it look like?



And how does it look like?



And how does it look like?

